Agentic AI for Vulnerability Confirmation and Patch Validation in Open-Source Software

SC4079

Dheeraj

College of Computing and Data Science
Nanyang Technological University

Supervisor: Assoc Prof. Chee Wei Tan

A Final Year Project Report
Academic Year 2026/2027

October 13, 2026

Abstract

An artificial intelligence system's ability to exploit software does not establish that it can reliably secure it. This project extended an existing exploit-confirmation prototype into a two-part, evidence-gated system for using large language models to secure open-source software: (1) dynamic confirmation and patch validation for already-published CVEs, and (2) reachability-filtered static analysis on real upstream free5GC source. Across five evolving pipeline rounds and a 36-run hosted-Claude comparison, dynamic exploit confirmation proved consistently achievable: all six catalogue classes were confirmed at least once, and every hosted model confirmed all six exploits. Patch validation proved substantially harder, and its own acceptance criterion needed correction twice. The original two-check criterion was found, by trace-level inspection, to accept a patch whose vulnerable route had merely crashed, indistinguishable to the automated check from a genuine fix; this was corrected into a four-outcome verdict. External review then found that corrected verdict itself still only checked for crash-absence on a benign request, not correct legitimate behaviour: a limitation this project reproduced and closed where possible.

Three independent real-upstream cases reached this project's strongest standards of evidence: a real, pinned-version PyPI package (`jaraco.context`, CVE-2026-23949); a real, disclosed, unauthenticated-disclosure advisory in OpenEMR (GHSA-q366-cv5v-83w8), confirmed by content-verified request/response evidence, transferring this method to a second domain, healthcare; and all four reachable handlers of a real, disclosed free5GC vulnerability (CVE-2026-40248): a harness running the actual vulnerable and patched `free5gc/udr` binaries inside a fuller deployment (a real MongoDB instance and a real free5GC NRF) confirmed, by direct HTTP exchange, that the vulnerable build permits an unauthorized read of a single record, an unauthorized read of the entire subscription collection, an unauthorized write, and an unauthorized delete, all through the same one-line missing-`return` defect, and that the real fix commit closes all four while preserving legitimate behaviour. This project's contribution is not a single confirmation or patch-acceptance rate, but an account of how far an evidence-gated workflow can progress from a suspected vulnerability to reproducible evidence, and where its own validation judgments can fail: a limitation this project discovered, corrected, and further narrowed after independent review.

Acknowledgements

I would like to thank my supervisor, Associate Professor Chee Wei Tan, for his guidance and feedback throughout this project. I also thank the Nanyang Technological University College of Computing and Data Science for the resources that supported this work, and the fellow student whose ProvTrail tool this project integrates with (Chapter 1) for that collaboration.

Contents

Abstract **i**

Acknowledgements **ii**

1 Introduction **1**

1.1 Background 1
1.2 Research Gap 2
1.3 Objectives 2
1.4 Scope 2
1.5 Project Schedule and Scope Evolution 3
1.6 Ethics and Responsible Use 4

2 Literature Review **5**

2.1 Overview 5
2.2 Agentic AI for Automated Security Testing 5
2.3 Cyber Reasoning Systems and the AIxCC/OSS-CRS Ecosystem 6
2.4 Reachability-Filtered Vulnerability Triage 7
2.5 Research Gap 9

3 Methodology **10**

3.1 Project Context 10
3.2 Methodology Overview 10
3.3 Alternative Approach Considered 12
3.4 System Architecture 12
3.4.1 System Description 12
3.4.2 Major System Components 13
3.4.3 Architecture Diagram 13
3.5 Data Sources and Preprocessing 16
3.5.1 Advisory and Source Data 16
3.5.2 Preprocessing Steps 16
3.6 Algorithms and Models Used 17

3.6.1 Model and Algorithm Selection . . . 17
3.6.2 Key Parameters and Settings . . . 18
3.7 Implementation Details . . . 18
3.7.1 Programming Environment . . . 18
3.7.2 Implementation Procedure . . . 19
3.8 Testing and Validation . . . 19
3.8.1 Testing Strategy . . . 19
3.8.2 free5GC Runtime Validation Harness . . . 20
3.9 Evaluation Metrics . . . 22
3.10 Limitations and Assumptions (Method-Related) . . . 24

4 Results and Discussion **26**
4.1 Results . . . 26
4.1.1 Exploit Confirmation Across the CVE Catalogue . . . 26
4.1.2 Patch Generation and Validation . . . 27
4.1.3 Real-Upstream Validation: CVE-2026-23949 . . . 33
4.1.4 Real-Upstream Validation: OpenEMR (GHSA-q366-cv5v-83w8) . . . 35
4.1.5 Multi-Model Comparison . . . 36
4.1.6 Multi-Model Comparison: Hosted Claude Models . . . 37
4.1.7 Reachability Filtering on Real Upstream Source . . . 40
4.1.8 Classification Ablation . . . 42
4.2 Discussion . . . 45
4.2.1 Interpreting the Confirmation-Rate Progression . . . 45
4.2.2 Why Patch Validation Lagged Confirmation . . . 46
4.2.3 Runtime and Model-Size Findings . . . 48
4.2.4 Classification Reliability and the Structured-Data Advantage . . . 48
4.2.5 Relationship to the AIxCC/OSS-CRS Benchmark Context . . . 49
4.2.6 Limitations, Trade-offs, and Sources of Uncertainty . . . 49
4.3 Summary and Implications . . . 50

5 Conclusion **52**
5.1 Summary of the Project . . . 52
5.2 Objective Closure . . . 52
5.3 Key Findings . . . 54
5.4 Contributions . . . 55
5.5 Limitations . . . 56
5.5.1 Final evidence freeze and interpretation . . . 57
5.6 Future Work . . . 58

A Reproducibility Details **63**
A.1 Full CVE Catalogue 63
A.2 free5GC Runtime Validation Evidence Bundle 64
A.3 Model and Sampling Configuration 67
A.4 Hosted Claude Comparison: Per-CVE Cost and Time 70
A.5 Validation Criteria (Quick Reference) 72
A.6 Relationship to AIxCC/OSS-CRS Vocabulary (Condensed) 73

B Selected Trace Excerpts **74**
B.1 The False-Acceptance Case: CVE-2026-78683, Round 4 vs. Round 5 . . . 74
B.2 The Markdown-Fence Crash (Round 3) 74
B.3 free5GC Upstream Compile Verification 75
B.4 free5GC LLM-Generated Patch: Multi-Model Comparison 75
B.5 free5GC Adversarial Bypass-Probe: Can a Model Find a Way Past the Real Fix? 77
B.6 Classification Ablation: NVD/GHSA Ground-Truth Cross-Check 80

C Verbatim Prompts (Representative Sample) **81**

D Repository and Artifact Access **83**

List of Figures

3.1 System architecture: advisory-driven exploit confirmation pipeline, from advisory intake through classification, dynamic execution, self-improvement, and patch generation/acceptance. 14
3.2 The Stage 3.6 interpreter branch: routes execution to `python` or `node` depending on target file extension. 15
3.3 The shared live-model backend used across Stages 3.5, 3.7, and 3.8, including the persisted-lesson store. 15

4.1 Dynamic confirmation rate across the five catalogue verification rounds (Table 4.1). The non-monotonic dip at Round 3 and the jump at Round 4 are discussed in Section 4.2.1. 27
4.2 Total measured cost against `confirmed_fix` count for each of the 6 hosted Claude models (Table 4.6). Cost and screening-pass rate are visibly not monotonically related: Haiku 4.5, the cheapest model, ties the best rate. 39

List of Tables

3.1 Major system components and their functions. 13
3.2 Advisory and source data used in evaluation. 16
3.3 Representative models and algorithms invoked by the pipeline (not the full eight-model comparison, which is reported in full in Table 4.5). 17
3.4 Key parameters and settings. 18
3.5 Programming environment. 18
3.6 Evaluation metrics and their justification. 22

4.1 Dynamic confirmation rate by round. 26
4.2 The six surviving patched artifacts under the stronger, content-verified reassessment: expected vs. actual response on both the malicious and legitimate request. 31
4.3 Real-upstream four-way validation result: CVE-2026-23949 (`jaraco.context`). 34
4.4 Real-upstream validation result: GHSA-q366-cv5v-83w8 (OpenEMR `admin.php`). 35
4.5 Confirmation rate by model (6 CVEs each). 37
4.6 Confirmation and patch outcomes by hosted Claude model (6 CVEs each, 36 runs total). 38
4.7 Stage 2 classification ablation (n=6). 43
4.8 Prose-only condition: micro-averaged precision, recall, and F1. 44
4.9 Prose-only condition: exact-CWE confusion matrix (n=6, one instance per true class). 45

5.1 Objective-by-objective closure. 53

A.1 All nine CVEs used across this project's evaluation: six in the main catalogue (Chapter 4), two in the free5GC work (Chapter 3), and one in the ProvTrail integration (Chapter 3). 63
A.2 free5GC runtime validation evidence bundle (`free5gc_runtime_case/`). . . 65
A.3 Exact configuration values used across the live-LLM evaluation (Chapter 4), read directly from `cve_pipeline.py`. 67
A.4 Round-to-commit mapping for the live-LLM catalogue runs (Chapter 4). 69

A.5 Per-CVE cost (USD) and wall-clock time (s) for every model–CVE cell in the hosted Claude comparison (36 cells total). 70

B.1 free5GC LLM-generated patch: eight locally-hosted models (Section B.4). 76
B.2 free5GC LLM-generated patch: eight hosted Claude models, real metered cost (Section B.4). 76
B.3 free5GC bypass-probe: eight locally-hosted models against the real fix (Section B.5). 78
B.4 free5GC bypass-probe: eleven named hosted Claude models against the real fix, real metered cost (Section B.5). 79
B.5 Ground-truth CWE cross-check between NVD and GitHub Security Advisories (Section 4.1.5). 80

D.1 Where each result reported in this thesis lives in the repository. 84

Chapter 1

Introduction

1.1 Background

An artificial intelligence (AI) system's ability to exploit software does not establish that it can reliably secure it. Recently in July 2026, OpenAI models undergoing cybersecurity evaluation with reduced safeguards bypassed isolation controls and compromised parts of Hugging Face's infrastructure [1]. The incident reminded us why advances in automated exploitation must be followed up with careful evaluation of their defensive use.

Agentic AI offers one approach to automated security testing. An agent uses a Large Language Model (LLM), which processes text and code in order to select tools as well as revise its actions using feedback [2]. For example, it would generate a test for a suspected vulnerability, run it, and investigate an unsuccessful attempt. Weiss and Khayet developed a pipeline that generates exploits from published vulnerability information and checks them against both vulnerable and patched software versions [3]. Their method formed the basis of AI-exploit-CVE, a prototype developed and presented at DEFCON Singapore. Weiss and Khayet's own account is a personal blog post, not a peer-reviewed result, and it is used accordingly throughout this project: as the origin of the idea being extended, never as evidence for a claim this project itself depends on. Every empirical result reported in this thesis (the confirmation and patch-acceptance rates, the real-upstream cases, the multi-model comparisons) is this project's own independently generated and dynamically confirmed evidence, not inherited from their report.

Broader capabilities have emerged from the Defense Advanced Research Projects Agency's AI Cyber Challenge (AIxCC), which discovered 54 of 63 synthetic vulnerabilities and patched 43, approximately 86% and 68% of the total [4]. Its winning system, ATLANTIS, combines language models with program analysis to discover vulnerabilities and generate fixes [5]. The Open Source Cyber Reasoning System framework (OSS-CRS) provides a common interface for building and running automated vulnerability discovery and fixing systems [6]. These developments provide a foundation for extending the

prototype beyond individual, disclosed vulnerabilities.

1.2 Research Gap

However, applying these capabilities requires evidence that the tests represent the software being assessed. An exploit that succeeds against a simplified reproduction may not work against the real application. Likewise, a patch that compiles may leave the vulnerability unresolved or disrupt legitimate functions. The question for this project is therefore how to extend the prototype to confirm suspected vulnerabilities and evaluate fixes in selected software targets.

1.3 Objectives

This study aims to develop and evaluate an agentic workflow for confirming disclosed vulnerabilities and validating candidate patches with executable evidence, using the stage vocabulary and evidence-gating philosophy of ATLANTIS and OSS-CRS as a conceptual frame rather than integrating their implementations directly (Chapter 2 states precisely where this project's stages align with theirs and where they do not). The workflow is evaluated on two tracks: a catalogue of six disclosed CVEs reproduced as minimal local targets, and free5GC, an open-source implementation of a 5G wireless network core used for research and testing [7], where one disclosed vulnerability is analysed and patched against the real, cloned upstream source rather than a reproduction, and evaluated as a real upstream runtime case: alongside compile-time verification of the broader four-handler patch set, the specific handler responsible for the disclosed CVE (CVE-2026-40248) was additionally run and exercised live, against a real local MongoDB instance, to confirm the exploit and the fix by direct HTTP exchange rather than by compilation alone. A further, narrower integration links the pipeline to ProvTrail, a companion student's tool for identifying JavaScript and TypeScript code resembling known vulnerable implementations; this integration is limited to SARIF-based ingestion plumbing and one hand-authored, dynamically confirmed demonstration case, not a systematic evaluation of ProvTrail's own flagged findings. SARIF, the Static Analysis Results Interchange Format [8], is the interchange format between ProvTrail's static findings and this project's dynamic-confirmation pipeline.

1.4 Scope

Evaluation covers the six-CVE reproduction catalogue described above, run across multiple verification rounds and across a comparison of eight locally hosted models plus a later

36-run comparison of six hosted Claude models, and the one free5GC case study, in controlled local environments; within that case study, all four analysed handlers (CVE-2026-40248) were additionally evaluated at runtime against a fuller deployment (a real, locally hosted MongoDB backend and a real free5GC NRF), rather than by compile-time verification alone, with OAuth2 enforcement explicitly disabled and out of scope for this evaluation. This assesses whether suspected vulnerabilities can be dynamically reproduced and whether a proposed patch causes the original exploit to fail under an automated criterion. Full functional preservation (whether a patch also preserves the affected route's behaviour for a legitimate, non-malicious request, verified against its exact expected content rather than merely the absence of a crash) was not part of the original broad synthetic evaluation across the full catalogue and every model compared; it was added later as a stronger, content-verified reassessment for six recoverable patched artifacts (Section 4.1.2) and for two real-upstream cases (Sections 4.1.3 and 4.1.4), not as a check performed uniformly across every result in this evaluation. Results obtained from simplified reproduction are distinguished from those obtained using actual software, with unsuccessful and inconclusive tests recorded. The findings are intended to help developers judge the evidence supporting an automated vulnerability report or proposed patch, including where that evidence, or the criterion used to produce it, was found to be incomplete.

1.5 Project Schedule and Scope Evolution

The project proceeded in four broad phases over the academic year: extending the inherited exploit-confirmation prototype with dynamic execution and self-improvement; building and correcting the patch-acceptance criterion described in Chapter 4; porting and applying the reachability-filtering capability to real free5GC source and extending it to runtime confirmation; and the later multi-model comparisons, real-upstream validation cases, and final evidence freeze. This is a chronological grouping of when work happened over the academic year, not the same four-way split as the Methodology Overview's "four main stages" (Section 3.2), which groups the same underlying work by pipeline component instead; the two do not and are not intended to map onto each other one-to-one. The work completed matches this final scope, but the scope itself was not the one originally proposed. The project's original framing described its goal as using fuzzing to secure open-source software with LLM agents. During implementation it became clear that no stage of the pipeline performs coverage-guided fuzzing or any other form of automated input mutation; every exploit and patch check in this project is instead generated and executed directly, without a fuzzing loop. Continuing to describe the project with that framing would have been inaccurate, so it was deliberately dropped, with the supervisor's agreement, in favour of the framing used throughout this report: LLM-assisted exploit confirmation

and patch validation for already-disclosed CVEs, together with reachability-filtered static analysis on real upstream source. This change affected only the project's stated framing, not its technical direction; the pipeline stages, evaluation targets, and evidence-gating approach described in this report were not restructured as a result, and no planned work was dropped to accommodate it.

Evaluation subsequently broadened to three further model-access methods beyond the Ollama and Claude CLI comparisons above (a direct Claude API gateway, the OpenAI API, and the Codex CLI, together exceeding fifty further model identities evaluated across the same free5GC, OpenEMR, and six-CVE-catalogue targets) summarised in Section 5.5.1 and reported in full in the project repository's `docs/RESULTS_BY_ACCESS_METHOD.md` and `docs/FINAL_EVIDENCE_SUMMARY.md`, rather than reproduced chapter-by-chapter here alongside the original Ollama/Claude CLI comparisons they extend.

1.6 Ethics and Responsible Use

All exploitation and patch testing in this project was carried out against locally hosted, self-controlled targets: either generated reproductions run on a private development machine, or a real upstream package (`jaraco.context`) and a real upstream repository (`free5gc/udr`) exercised entirely within isolated local environments the author controlled, never against a public or third-party service. No exploit generated by this project was deployed against, or tested on, any system outside the author's own local infrastructure, and no live production service, public deployment, or third party's data was accessed, probed, or affected at any stage. Every CVE and CWE studied is a real, already-published, publicly disclosed advisory (verified against NVD, GHSA, and the relevant upstream project's own advisory or fix commit before use, per Appendix A); this project did not discover, disclose, or otherwise handle any previously unknown vulnerability, so no responsible-disclosure obligation to a vendor arose during this work. Local HTTP servers used to serve crafted archives or payloads (for example, the real-upstream `jaraco.context` case, Section 4.1.3) were bound to `127.0.0.1` only and never exposed beyond the local machine. Should this project's methodology be extended to a previously undisclosed finding in future work, the author commits to following standard coordinated/responsible disclosure practice (privately notifying the affected project or vendor, allowing a reasonable remediation window, and withholding public detail until a fix is available) rather than the immediate, open reporting appropriate for the already-published CVEs studied here.

Chapter 2

Literature Review

2.1 Overview

This project sits at the intersection of three bodies of work: agentic AI systems that use large language models (LLMs) to carry out multi-step security tasks, Cyber Reasoning Systems (CRSs) that automate vulnerability discovery and repair at scale, and reachability-based static analysis techniques that narrow the scope of what such systems must examine. This chapter reviews each in turn, moves from isolated per-system description toward the themes that connect them, and closes by identifying the specific gap (evidence-gated confirmation of both exploits and patches against representative targets) that motivates this project's objectives, introduced in Chapter 1.

2.2 Agentic AI for Automated Security Testing

An agent, in the sense used throughout this literature, is a system in which an LLM processes text and code, selects among available tools, and revises its own actions based on feedback from their results, rather than following a fixed script [2]. This feedback loop is what distinguishes an agentic approach from a single-shot LLM query: an agent that generates a candidate exploit can, in principle, run it, observe whether it succeeded, and try again with the failure in hand.

Fang, Bindu, Gupta, and Kang established the underlying capability this loop depends on: given only a CVE's advisory text, GPT-4 autonomously exploited 87% of a benchmark of 15 real one-day vulnerabilities, against 0% for every other model tested and for the ZAP and Metasploit scanners, with success falling to 7% once the advisory text was withheld: showing that advisory text, not general-purpose exploitation skill, was doing most of the work [15]. A follow-up scaled this same result from a single agent on known one-day vulnerabilities to teams of agents on genuine zero-days selected to postdate the evaluated model's training cutoff, reporting a 4.3× improvement over prior

single-agent frameworks across a 14-vulnerability benchmark via a hierarchical planning-and-subagent architecture [18]. Weiss and Khayet demonstrated this same loop concretely for vulnerability exploitation, showing that an agent could take a published CVE's advisory text and generate a working exploit against it in under fifteen minutes [3]. Their pipeline formed the direct basis for AI-exploit-CVE, the prototype this project extends. At a larger scale, CVE-GENIE's multi-agent reproduction framework reproduced 428 of 841 (51%) real 2024–2025 CVEs with verifiable exploits at an average cost of $2.77 per CVE [19], giving a population-level success rate and cost baseline against which this project's own much smaller, fully manually-audited six-CVE catalogue (Chapter 4) can be read. What Weiss and Khayet's work established, and what it left open, both matter for how this project is scoped: they showed that advisory-to-exploit generation is tractable for an agent, but their evaluation, like much of the broader "AI can hack" literature this project's Introduction opens with [1], centres on the offensive capability itself rather than on how confidently a defender could trust the result. An exploit that an agent reports as successful and an exploit that has actually been shown to work against a representative target are not the same claim, and the literature reviewed here does not consistently distinguish between them. PoC-Gym quantifies exactly this gap for Java CVEs: of 116 LLM-generated candidates that passed its own runtime validation (completed execution and produced a success signal), only 65 also passed post-hoc verification against the actual vulnerable code location: meaning 44% of runtime-successful candidates were false positives that never reached the target flaw [14]. This project's own dynamic-confirmation stage (Section 3.7.2) is designed against exactly this risk, replacing an earlier design where generated exploits were written to disk but never actually run with one that checks the exploit script's real process exit code; PoC-Gym's result is a reminder that this choice narrows the gap rather than closes it outright.

2.3 Cyber Reasoning Systems and the AIxCC/OSS-CRS Ecosystem

A Cyber Reasoning System generalises the single-exploit agentic loop into a full pipeline: discover an unknown vulnerability in a codebase, triage it, generate a proof of vulnerability, and generate and validate a patch, all without a human specifying the bug in advance. DARPA's AI Cyber Challenge (AIxCC) evaluated seven finalist CRSs (Atlantis, Buttercup, RoboDuck, FuzzingBrain, Bug Buster, Artiphishell, and Lacrosse) against a shared benchmark of synthetic vulnerabilities seeded into real open-source projects, with the finalists collectively discovering 54 of 63 seeded bugs (86%) and patching 43 (68%), per DARPA's own results announcement and, independently, the first systematic peer-reviewed analysis of the competition's design, architectures, and results [4, 17]. The winning system,

ATLANTIS, combined LLM-driven reasoning with traditional program-analysis techniques across its discovery, triage, and patch-generation stages [5], and the Open Source Cyber Reasoning System project (OSS-CRS) subsequently generalised several AIxCC-era systems into a common runtime so they could be pointed at real OSS-Fuzz projects rather than only the competition's own challenge problems [6].

Two things follow from reading AIxCC and OSS-CRS alongside Weiss and Khayet's narrower exploit-generation work, stated as an observation and a hypothesis rather than an established finding. AIxCC's aggregate 86%-discovery/68%-patch split, reported across seven finalist systems on a large competition benchmark, is not directly comparable in task, scale, or denominator to this project's own six-CVE confirmation/patch-validation results (Section 4.2.5), and this project's own patch-validation numbers have themselves been shown, by trace-level inspection, to include at least one false acceptance (Chapter 4): so neither figure should be read as an independently reliable measurement to compare against the other. What both bodies of work do suggest, more cautiously, is that generating a working patch may plausibly be harder than discovering or confirming the underlying bug across genuinely different systems and scales; this project's own results are consistent with that hypothesis without independently proving it holds as a general property of the task. Second, AIxCC and OSS-CRS finalists are primarily full discovery systems (they find vulnerabilities nobody has previously identified) though this is not absolute: FuzzingBrain, one AIxCC finalist, includes a documented mode for assessing externally supplied findings (for example, from static-analysis tools) rather than only its own discoveries [10], which is closer in spirit to a known-finding confirmation task than a pure discovery system. This project's own scope remains narrower still and more specific: it confirms and patches already-published CVE advisories specifically, with dynamic execution evidence for every case, rather than triaging static findings of unstated provenance. Within the small, non-systematic set of sources reviewed for this chapter, this specific combination (disclosed-CVE confirmation via real subprocess/HTTP execution, evaluated with the same evidence-gating discipline AIxCC-class systems apply to their own outputs) is not something this review found duplicated elsewhere; this is stated as a gap in the material reviewed, not a claim about the full field.

2.4 Reachability-Filtered Vulnerability Triage

A separate line of work addresses a different bottleneck: even once a codebase is the analysis target, an LLM or static analyser cannot practically reason over every function in a large real-world project. OpenAnt addresses this by decomposing a codebase into a call graph and performing breadth-first reachability analysis from attacker-controlled entry points, reporting scope reductions of up to 97% on real projects including OpenSSL, WordPress, and Flowise, before applying LLM-driven adversarial verification and sandboxed dynamic

testing to the reduced candidate set [9]. An independently reported AIxCC finalist system, FuzzingBrain [10] (competition repository name `afc-crs-all-you-need-is-a-fuzzing-brain`), converges on a similar three-stage shape (static reachability partitioning, LLM-based "suspicious point" triage, then fuzzing-driven verification) arrived at separately from OpenAnt's own design.

Read together, OpenAnt's and FuzzingBrain's independently-arrived-at similarity is at least suggestive that reachability-first filtering is not an idiosyncratic design choice specific to one system, though this observation rests on the small, non-systematic set of sources reviewed for this chapter rather than a comprehensive survey of the field, and should be read with that scope in mind. Patch validation rigour in this literature varies: FuzzingBrain's own published validation criteria require a patch to compile successfully and to pass a supplied regression-test suite before being accepted [10], which is at least as strong as, and in the regression-testing respect stronger than, this project's own two-check acceptance criterion (Section 4.1.2). This gap between a patch passing validation and a patch actually being correct is not new to LLM-based repair: Qi, Long, Achour, and Rinard's decade-old analysis of three classical generate-and-validate repair systems (GenProg, RSRepair, and AE) found that errors in the validation infrastructure itself meant most reported patches were not even plausible under their own test suites, and that almost all patches judged plausible were equivalent to deleting the relevant functionality rather than fixing it: establishing that passing a validation suite and being a correct patch have never been the same claim, well before LLMs entered the picture [16]. Two further, more recent sources quantify this same validation-rigour gap directly rather than only arguing for it: VulnRepairEval found that exploit-based differential testing (requiring the original PoC to fail against the patched code, not merely that the patch applies) reveals a severe performance deficit hidden by superficial validation: even the best of twelve evaluated LLMs solved only 5 of 23 real-world Python CVEs under this stricter standard [12]. PVBench goes further still, constructing developer-authored "PoC+" tests that encode root-cause and specification intent beyond mere exploit-blocking, and finding that over 40% of patches three state-of-the-art automated-repair systems had validated as correct under basic testing failed this stronger standard [13]. This project's own round-5 false-acceptance finding (Section 4.2.2) (a patch accepted by a basic two-check criterion that trace-level inspection later showed was an unrelated crash, not a deliberate fix) is a single, concrete instance of exactly the overestimation pattern both of these benchmarks measure at scale. What distinguishes this project's own reachability work (Chapter 3) is not a general gap in the literature's rigour, but a specific combination: applying reachability-filtered triage to an already-disclosed CVE and verifying the resulting patch by cloning the actual upstream repository and rebuilding it against that project's real, unmodified dependency graph, rather than against a synthetic or sandboxed target. This project's free5GC case study demonstrates that specific combination for one real

vulnerability, detailed further in Chapter 3; it is offered as this project's own contribution, not as evidence that the wider literature lacks compile-time verification altogether.

2.5 Research Gap

Three observations from the small set of sources reviewed above, taken together, motivate this project's objectives; they are offered as a rationale drawn from a limited, non-systematic review, not as a claim about the full field. First, agentic exploit generation (Weiss and Khayet, and the broader capability the Hugging Face incident illustrates [1]) has been demonstrated as tractable, but the evaluations reviewed here do not consistently separate "the agent reports success" from "the exploit was dynamically confirmed against a representative target": a distinction this project treats as a first-class requirement rather than an afterthought. Second, AIxCC-class systems are primarily oriented toward discovering previously unknown vulnerabilities at competition scale (with FuzzingBrain's external-finding assessment mode a partial exception); this project instead studies a narrower, everyday situation (triaging a dependency's already-disclosed CVE) with a lighter-weight, dynamic-evidence workflow scoped specifically to that task, without claiming from the small set of sources reviewed here that this situation is underserved by the field as a whole. Third, reachability-filtered triage (OpenAnt, FuzzingBrain) is established as an effective scope-reduction technique with, in FuzzingBrain's case, genuinely rigorous compile-and-regression-test patch validation; what remains specific to this project is combining that filtering technique with compile verification against a real, live-cloned upstream project's dependency graph for an already-disclosed CVE, rather than a claim that the wider literature omits compile-time verification.

This project studies the following combination under its own stated conditions, distinct from each source reviewed above in a specific, statable way: unlike Weiss and Khayet's exploit-generation work, it extends generation with dynamic execution evidence and a two-check patch-acceptance criterion, evaluated with real subprocess and HTTP evidence rather than static description (Chapter 3); unlike OpenAnt's published evaluation, it verifies a reachability-filtered patch by cloning and rebuilding the actual upstream repository rather than an isolated file or sandbox (Chapter 3); and unlike the AIxCC-aggregate figures, it reports its own exploit-confirmation and patch-acceptance rates across multiple models and multiple verification rounds on a small, fully described catalogue (Chapter 4), so its own discovery/patch pattern can be inspected directly rather than assumed to match AIxCC's aggregate. Findings for JavaScript and TypeScript, via the ProvTrail integration introduced in Chapter 1, extend this same evidence-gated approach to a second ecosystem, using SARIF [8] as the interchange format between ProvTrail's static findings and this project's dynamic-confirmation pipeline.

Chapter 3

Methodology

3.1 Project Context

This project extends an existing single-vulnerability exploit-confirmation prototype, AI-exploit-CVE, into a two-part evidence-gated system for using large language models (LLMs) to secure open-source software: (1) LLM-assisted exploit confirmation and patch validation for already-published CVEs, and (2) reachability-filtered static analysis on real upstream source, applied to a disclosed vulnerability in free5GC, an open-source 5G core network implementation [7]. ATLANTIS and OSS-CRS (Chapter 2) inform this project's stage vocabulary and evidence-gating philosophy rather than being integrated as code: no component of ATLANTIS or OSS-CRS is imported, called, or run by this project, and Appendix A.6 states explicitly where the analogy between this project's stages and theirs holds and where it does not. A secondary integration links the pipeline to ProvTrail, a companion project that statically identifies JavaScript/TypeScript code resembling known-vulnerable implementations; this integration currently consists of SARIF ingestion plumbing (`provtrail_bridge.py`) and one hand-authored, dynamically confirmed JavaScript case (CVE-2024-48910, Prototype Pollution), not a systematic evaluation of ProvTrail's own flagged findings against real scanned projects, and this narrower scope is stated here rather than implied by the architecture description alone. The system falls under the Software System/Application category, with a Data Analytics component in its reachability-filtering and metrics stages.

3.2 Methodology Overview

This project was carried out in four main stages. First, the existing CVE-driven exploit-confirmation pipeline (advisory intake, vulnerability classification, exploit-artifact generation, dynamic execution, self-improvement on failure, and patch generation with regression validation) was extended and hardened against a catalogue of six published CVEs spanning

six vulnerability classes. Next, a reachability-filtering engine (ported from an earlier, standalone prototype developed for the same project before its scope was redirected) was applied to a real, disclosed CVE in free5GC, producing an AST-based call graph reduction and a patch verified to compile against the actual upstream dependency graph. This was followed by the integration of ProvTrail's SARIF/AI-text export as an additional advisory source, allowing the pipeline to be driven by statically located JavaScript/TypeScript findings rather than only by NVD/GHSA feed entries. Finally, the system was evaluated across multiple rounds of live, locally hosted LLM verification (using Ollama-served open-weight models [11]), during which two implementation defects were identified, fixed, and re-verified, and the pipeline's evidentiary claims were independently confirmed to hold under live model generation, not only under stubbed responses.

High-level workflow

1. Advisory intake: fetch a CVE/GHSA advisory, or consume a ProvTrail-flagged finding, as the unit of work.

2. Vulnerability classification: categorise the advisory into one of six classes (SSRF, SQL injection, OS command injection, XSS, path traversal, deserialisation) or route it to the reachability engine for memory-safety-class C/C++/Go targets.

3. Exploit-artifact generation: synthesise a proof-of-concept and a minimal reproduction target from the advisory's described bug pattern.

4. Dynamic execution: run the generated artifact against the reproduction target and record a real process exit code / HTTP response as evidence.

5. Self-improvement: on a genuine, executed failure, attempt a fix via a persisted lesson, then an LLM-generated revision, then a rule-based fallback, bounded to three iterations.

6. Patch generation and validation: generate a candidate patch and accept it only if the original exploit now fails against it and the target's health check still passes.

7. Reachability analysis (parallel track): parse real upstream source into an AST-derived call graph, perform breadth-first reachability from stated entry points, and generate and compile-verify a patch against the real dependency graph.

8. Reporting and metrics: aggregate per-CVE outcomes into structured reports and summary metrics (confirmation rate, patch-acceptance rate, reachability reduction, wall-clock time, LLM call count).

3.3 Alternative Approach Considered

Before continuing with the existing advisory-driven architecture, a proposed redesign was evaluated and deliberately not adopted: replacing CVE-driven exploit confirmation with a pipeline in which an LLM reads raw source code with no prior knowledge of any specific advisory, hypothesises a vulnerability from scratch, synthesises a triggering input purely by reasoning rather than by search or mutation, and validates the hypothesis by replaying that input once against an instrumented build. This alternative was rejected for four reasons. First, its own proposed validation plan (a single smoke test against one example file) was judged insufficient grounds to commit to restructuring the project's architecture before that assumption was independently tested. Second, this project's existing exploit-artifact generation is always guided by an advisory's own stated root cause, a materially easier task than zero-shot vulnerability hypothesis generation with no prior knowledge, so this project's own results do not transfer as evidence that the alternative's harder, blind variant would work. Third, the proposal would have introduced a second, disconnected system rather than extending the existing, already-validated pipeline. Fourth, it required setting up new local model infrastructure for a speculative direction while the existing, evidence-gated pipeline was already in a documented, working state. This alternative remains available to revisit as a bounded, separately-evaluated extension, but was not adopted as the basis for this project.

3.4 System Architecture

3.4.1 System Description

The system consists of two loosely coupled subsystems sharing a common advisory-driven design philosophy. The first, the main exploit-confirmation pipeline, is orchestrated by `cve_pipeline.py` and driven by a feed poller (`cve_watcher.py`) that dispatches pipeline runs for newly observed advisories from NVD, GHSA, or ProvTrail. The second, the reachability-filtering engine (`src/reachability/`), operates independently on real upstream source rather than on advisory text, and is invoked directly against a stated set of entry-point functions. A supporting bridge module (`provtrail_bridge.py`) converts ProvTrail's SARIF output into advisories the main pipeline can consume, and two standalone dynamic-exploitation labs (`ssrf_lab/`, `free5gc_lab/`) provide locally hosted, self-controlled targets for live confirmation where a generic Flask stand-in would not suffice.

3.4.2 Major System Components

Table 3.1: Major system components and their functions.

Component / Module	Function
`cve_pipeline.py`	Orchestrates advisory fetch, classification, artifact generation, dynamic execution, self-improvement, and patch generation/validation for a single CVE.
`cve_watcher.py`	Polls NVD, GHSA, and ProvTrail feeds on a fixed interval and dispatches pipeline runs across parallel workers.
`provtrail_bridge.py`	Converts ProvTrail's SARIF/AI-text export into advisories consumable by the main pipeline, falling back to NVD/GHSA when no scan is available.
`src/reachability/`	Performs tree-sitter-based AST parsing, call-graph construction, breadth-first reachability filtering, adversarial triage, and rule-based or LLM-based patch generation on real upstream source.
`src/pipeline/patch.py`	Generates a candidate patch for a confirmed exploit and validates it via a two-check regression test.
`src/metrics.py`	Aggregates confirmation rate, patch-acceptance rate, wall-clock time, and LLM call count across the report catalogue.
`free5gc_lab/`	A dependency-free Go reproduction of a disclosed free5GC vulnerability, dynamically exploited and patched in isolation from the main pipeline.
`ssrf_lab/`	A local FastAPI target providing genuine live-traffic confirmation for the SSRF vulnerability class.

3.4.3 Architecture Diagram

A system architecture diagram is included as Figure 3.1 below (landscape orientation, for legibility), illustrating the flow from advisory intake through classification, artifact generation, dynamic execution, self-improvement, and patch validation. Figures 3.2 and 3.3 detail two supporting mechanisms: the interpreter branch that routes Stage 3.6 execution to `python` or `node` depending on target file extension, and the shared live-model backend used by Stages 3.5, 3.7, and 3.8. All three diagrams are rendered directly from this project's own `architecture.html` source, not redrawn.

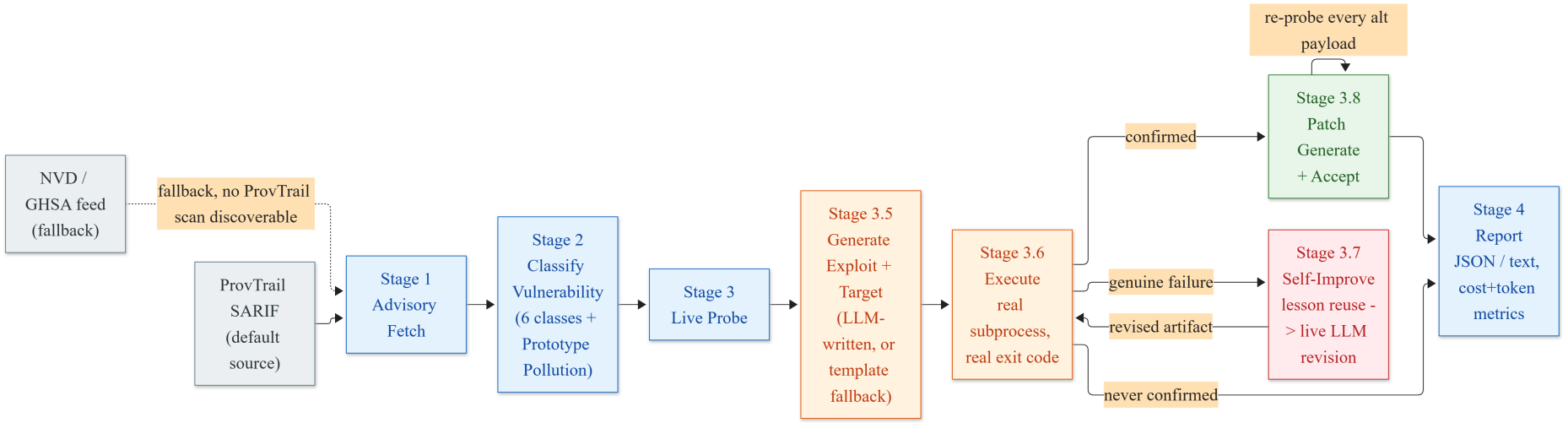


Figure 3.1: System architecture: advisory-driven exploit confirmation pipeline, from advisory intake through classification, dynamic execution, self-improvement, and patch generation/acceptance.

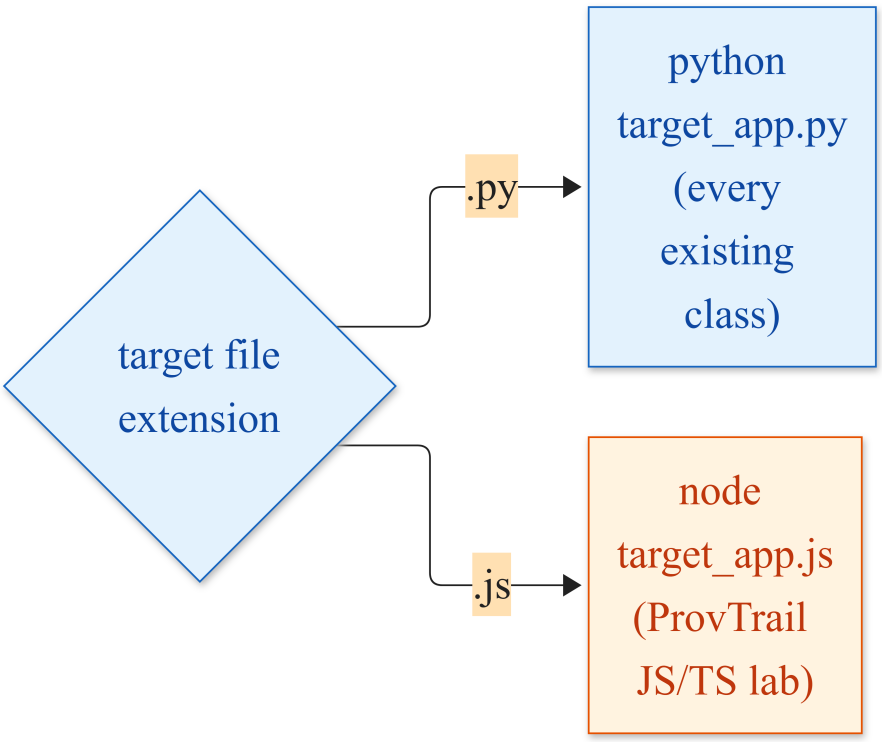


Figure 3.2: The Stage 3.6 interpreter branch: routes execution to `python` or `node` depending on target file extension.

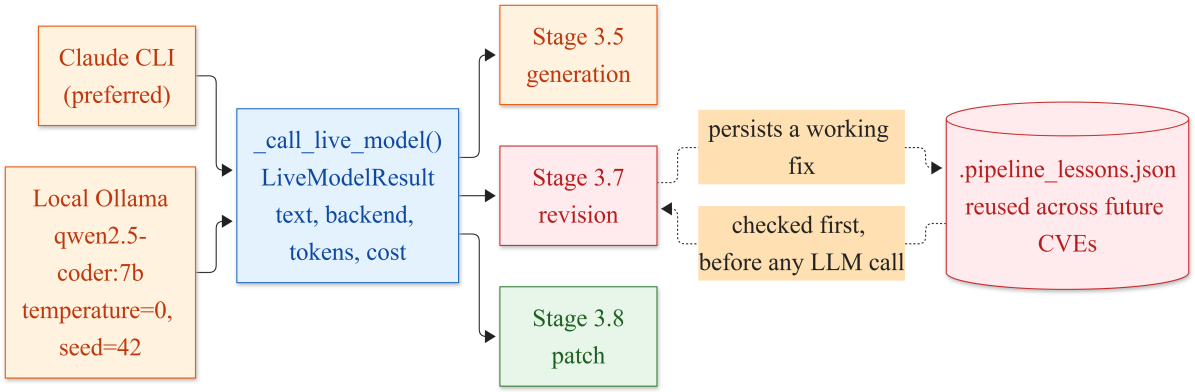


Figure 3.3: The shared live-model backend used across Stages 3.5, 3.7, and 3.8, including the persisted-lesson store.

3.5 Data Sources and Preprocessing

3.5.1 Advisory and Source Data

Table 3.2: Advisory and source data used in evaluation.

Item	Description
Advisory sources	NVD API v2, the GHSA feed, and ProvTrail's SARIF/AI-text export
Advisory volume (main catalogue)	Six CVEs, spanning six vulnerability classes (SQL injection, OS command injection, XSS, path traversal, deserialisation, SSRF)
free5GC: CVE-2026-40246	Evaluated separately as a hand-written Go reproduction (`free5gc_lab/`), dynamically exploited and patched at runtime; no fix commit documented for this CVE in this project's own records
free5GC: CVE-2026-40248	Evaluated against the real, cloned `free5gc/udr` upstream source at four selected handlers, with a documented fix commit and a build check at the pre-fix revision for all four; all four handlers additionally run at runtime against a fuller deployment (a real MongoDB instance and a real free5GC NRF, OAuth2 disabled) (Section 3.8.2)
Upstream source data	Real, verbatim free5GC/UDR source (`internal/sbi/api_datarepository.go`), fetched at the commit before and the commit of the disclosed fix, approximately 86 KB, 102 parseable functions

3.5.2 Preprocessing Steps

Advisory text was parsed to extract the vulnerability class, affected package, and any code-diff snippet included in the advisory. Where a code diff was present, it was used for static pattern matching during classification. For the reachability track, the fetched Go source file was parsed using a tree-sitter grammar into an abstract syntax tree, from which function definitions and call edges were extracted to construct a directed call graph. A stated list of four real HTTP entry-point handlers was then used as the seed set for breadth-first reachability traversal.

3.6 Algorithms and Models Used

3.6.1 Model and Algorithm Selection

Table 3.3: Representative models and algorithms invoked by the pipeline (not the full eight-model comparison, which is reported in full in Table 4.5).

Model / Algorithm	**Purpose**
Claude CLI / pydantic-ai agent backend	Advisory analysis, exploit-artifact generation, and patch generation when a hosted API or CLI session is available
Ollama-served `qwen2.5-coder:7b` (local, no API key)	Live-model verification of exploit-artifact generation and self-improvement revision
Ollama-served `qwen2.5-coder:1.5b` and `mistral:7b` (local)	Comparative live-model backends evaluated across verification rounds
Rule-based heuristics (e.g. `find_missing_return_after_response`)	Deterministic, offline fallback patch generation requiring no LLM call
Tree-sitter AST parsing + breadth-first search	Reachability filtering: construction of a call graph and traversal from stated entry points

3.6.2 Key Parameters and Settings

Table 3.4: Key parameters and settings.

Parameter	Value	Reason
Self-improvement iteration bound	3	Prevents unbounded LLM cost or looping past a signature with no known fix
Feed poll interval	60 minutes	Balances advisory freshness against feed rate limits
Parallel workers	2	Bounds concurrent LLM/API load while allowing cross-CVE parallelism
Reachability entry-point set	4 handlers	Reflects the real, externally reachable surface of the analysed free5GC module

3.7 Implementation Details

3.7.1 Programming Environment

Table 3.5 summarises the languages, libraries, and hardware the system was implemented and evaluated with.

Table 3.5: Programming environment.

Item	Details
Programming languages	Python (pipeline orchestration, pydantic data models, reachability engine); Go (free5GC target reproduction and the real upstream module under test)
Libraries / frameworks	pydantic-ai (agent orchestration), tree-sitter (AST parsing), Flask (main-catalogue reproduction targets), FastAPI (`ssrf_lab/`), Go's standard-library `net/http` (dependency-free lab server)
Hardware	13th Gen Intel Core i9-13980HX, 16 GB RAM, NVIDIA GeForce RTX 4060 Laptop GPU (Ollama-served models run on this GPU; no dedicated accelerator was required for the model classes evaluated); Windows 11 Home, build 10.0.26200

3.7.2 Implementation Procedure

The system was implemented incrementally, in the following order. First, the pre-existing exploit-confirmation pipeline was extended with dynamic execution, replacing an earlier design in which generated exploit artifacts were written to disk but never actually run. Self-improvement was then added as a bounded retry loop consulting, in order, a persisted-lesson store, an LLM-generated revision, and a rule-based fallback. Patch generation and validation was implemented next, gated on two simultaneous conditions: the original exploit failing against the patched target, and the target's health check continuing to pass. The reachability engine was subsequently ported from an earlier standalone prototype and re-validated against its own prior results before being applied to real free5GC source, first as an isolated file parse and later against the actual cloned upstream repository to confirm compile-compatibility with its full dependency graph. The ProvTrail bridge and the free5GC dynamic-exploitation lab were implemented as independent, loosely coupled additions, following the same dual vulnerable/patched-mode convention already established elsewhere in the system. Finally, the system was run across successive live-model verification rounds; two defects surfaced during this process (an unstripped markdown code fence in model-generated output, and stale execution-state fields left over from a prior run) were each isolated with a standalone targeted test, fixed, and the full catalogue was re-run to confirm the fix.

3.8 Testing and Validation

3.8.1 Testing Strategy

The system was tested by executing every generated exploit artifact as a real subprocess against a real, locally deployed target and recording its actual exit code or HTTP response, rather than treating generation as evidence of exploitability by itself. Patch validation was tested by re-running the original, unmodified exploit against the patched target and independently confirming a separate `/health` endpoint still responded correctly, so that a patch which crashed the target process entirely would be rejected. This check does not confirm that the vulnerable route itself still behaves correctly for a legitimate request, only that the application as a whole is still running; Chapter 4 reports a case where this specific gap allowed a patch containing an unrelated crash bug to be marked validated. For the reachability track, the generated patch was tested for compile-compatibility by cloning the actual upstream repository at the pre-fix commit, applying the patch to the real source file, and rebuilding against the module's genuine dependency graph. All four handlers within that track, CVE-2026-40248, were tested one tier further, at runtime, using a purpose-built harness (first a minimal local stand-in, then a fuller Docker-based

deployment with a real NRF) rather than compilation alone.

3.8.2 free5GC Runtime Validation Harness

The real, cloned `free5gc/udr` module was built twice from source, once at the vulnerable commit and once at its immediate successor, the real upstream fix commit. Both builds require a reachable MongoDB instance to start; a native MongoDB Community Server instance, run locally rather than inside a container, was used, since the experiment's only real infrastructure dependency was UDR's own database connection, not full containerisation. Both builds also require a reachable Network Repository Function (NRF) to complete their own startup: UDR's registration call to the NRF blocks its HTTP server from starting at all if it never succeeds, an infinite retry with no timeout, so a minimal stand-in NRF, answering only the one registration call UDR needs and asserting no broader NF-discovery capability, was required purely to let each build reach the point of serving real HTTP traffic. This stand-in had itself to be implemented as a small compiled Go program communicating over cleartext HTTP/2 with prior knowledge (h2c), rather than a simpler HTTP/1.1 script, because free5GC's own NRF client negotiates HTTP/2 unconditionally and rejects an HTTP/1.1 responder outright; this was discovered empirically; an initial HTTP/1.1 stand-in failed with a client-side framing error rather than a connection error, and was replaced once the cause was identified. With both dependencies satisfied, each build was driven through an identical request sequence over real HTTP: seed a subscription record, issue the CVE's malicious delete request, check whether the record still exists, reseed, issue a legitimate delete request for the same record, and check again: with every request, response, and process log captured as part of the resulting evidence bundle rather than only the final pass/fail outcome.

This minimal harness was subsequently extended to a fuller deployment and to the three remaining reachable handlers, using the official `free5gc-compose` Docker stack rather than the hand-written stand-in above: a real, official MongoDB image and a real, official free5GC NRF image (not a minimal stub), with a custom-built UDR image compiled from the same pinned vulnerable and fixed commits. OAuth2 enforcement, which the real NRF enables by default and which would otherwise require a full OAuth2 client-credentials flow unrelated to the vulnerability under test, was explicitly disabled in the NRF's configuration for this harness; this is a real, supported free5GC configuration option, not a synthetic bypass, but it means this result says nothing about whether OAuth2 enforcement would affect an attacker's ability to reach these handlers in a deployment that has it enabled. Under this fuller deployment, all four of the reachability engine's originally-flagged handlers (not only the DELETE handler) were driven through the same seed/malicious-request/benign-request pattern, again with every request, response, and container log captured.

This project does not use a conventional training/testing split or k-fold cross-validation, as it is not a statistical learning system being evaluated for generalisation; alternative validation designs considered included purely static rule-checking (rejected, as it cannot demonstrate an exploit actually succeeds) and reliance on a single verification run (rejected in favour of repeated rounds, since two genuine implementation defects were only surfaced by re-running the full catalogue after each fix). Instead, validation proceeded through repeated verification rounds across the CVE catalogue (five successive rounds of live-model runs) with each round's failures traced to a specific, named cause before the next round was attempted, rather than accepted as unexplained noise.

3.9 Evaluation Metrics

Table 3.6: Evaluation metrics and their justification.

Metric	**What It Measures**
Confirmation rate	Proportion of catalogued CVEs whose described bug pattern was dynamically confirmed exploitable (5 of 6)
Two-check patch acceptance rate	Proportion of confirmed exploits for which a generated patch met both conditions of the pipeline's own `patch_validated` field (the original PoC no longer succeeds, and a separate health endpoint still responds) first achieved in round 4, after zero acceptances across three prior rounds; this metric does not measure benign-route preservation or repair correctness, and is superseded as this project's headline patch metric by the corrected `patch_verdict` field (Section 4.1.2), retained here and throughout for continuity with earlier rounds
Reachability reduction	Percentage reduction in functions requiring analysis after BFS filtering (96.1%, 102 reduced to 4)
Wall-clock time per CVE	Runtime cost of a full pipeline run, as a practicality indicator
LLM call count, wall-clock time, and recorded API cost (USD)	Real, per-call figures read directly from the Claude CLI's or Ollama's own token/duration fields (not estimated). Every Ollama-backend call in this evaluation records $0.0000, a genuine zero *billed API charge* from local, unmetered inference, not zero underlying compute cost, which was not separately measured. Claude CLI calls initially recorded no cost figure at all, since the CLI's plain-text output mode exposes no usage metadata; switching to its structured JSON output mode (Section 4.1.6) exposes a real, non-zero, provider-reported `total_cost_usd` per call, used directly in Table 4.6

These metrics were selected to mirror the vocabulary used by DARPA's AI Cyber Challenge and the OpenSSF's OSS-CRS project to report cyber-reasoning-system results (bugs confirmed against bugs attempted, proof-of-vulnerability confirmation rate, and patch-acceptance rate) so that this project's results can be read against the same benchmark context introduced in Chapter 1, rather than against an ad hoc scale.

To keep these terms precise across the results reported in Chapter 4, evidence tiers are distinguished explicitly rather than collapsed into a single pass/fail label. **Dynamically confirmed** means a real subprocess exit code from a real HTTP exchange, or a real compiled-harness run, not a static description. **Patch accepted by the two-check criterion** means the original, unmodified proof-of-concept no longer meets its success condition against the patched local target, while a separate `/health` request still succeeds. This checks exactly one attack input and application liveness; it does not establish that the affected route handles a legitimate request correctly, that other attack inputs are blocked, or that the named upstream package has been repaired. This project does not use the term "plausible patch" or claim "regression-safety" for this flag: Chapter 4 reports a specific case (CVE-2026-78683, round 5) where the tested exploit request crashed the patched route with an unrelated import error, and this satisfied the exact criterion above, which is why a stronger label would overstate what it proves. This criterion is also weaker than the patch-validation bar used by at least one real AIxCC finalist system, FuzzingBrain, whose own published criteria require a patch to compile, eliminate all known proofs of the vulnerability, and pass a supplied regression-test suite [10]: a three-part bar this project's own two-check criterion does not match. A further tier, **correct** patch (one independently checked against the real upstream fix's own intent, or against benign-request and multi-payload behaviour, rather than only against the one exploit that triggered it) is, at this stage of the pipeline, not yet measured by this project; no diff-similarity scoring or second-reader adjudication against the real upstream fix's own intent is performed anywhere in this project, and this project's own patch-validation investigation (Section 4.1.2) shows why that distinction matters in practice. The benign-request half of this gap is later closed, for a bounded set of cases, not left purely as a limitation: Section 4.1.2's content-verified reassessment establishes exact expected benign-path behaviour for 6 recoverable patched artifacts, and Sections 4.1.3 and 4.1.4 establish it for the real-upstream maintainers' own fixes in three further cases; the strict diff-similarity/second-reader sense of "correct" remains unmeasured throughout. **Compile-verified** describes the free5GC reachability case study's original compile-only evidence and means `go build` succeeds against the real, live-cloned upstream module with the patch applied: explicitly weaker evidence than dynamic confirmation, and never reported in the same column without this qualification. All four of that case study's reachable handlers (CVE-2026-40248) were subsequently **runtime-confirmed**: the real, cloned module was built, run, and exercised over real HTTP for both the vulnerable and fixed commits, first against a minimal MongoDB-only

stand-in (Section 3.8.2) for the DELETE handler, then against a fuller deployment (real MongoDB and a real free5GC NRF, OAuth2 disabled) for all four handlers, meeting the same **dynamically confirmed** bar defined above rather than the weaker compile-verified one.

3.10 Limitations and Assumptions (Method-Related)

The methodology assumes that a faithful reproduction of an advisory's described bug pattern is representative evidence of that bug class, even where the reproduction target is a minimal stand-in rather than the actual named upstream package. For the six main-catalogue CVEs, the exploited target is a generated Flask application reproducing the described pattern; a confirmed exploit therefore demonstrates that the described pattern is exploitable as written, not that the real, named package is exploitable when actually installed and run. The live dynamic probe is genuine only for the SSRF class; the remaining five classes rely on static pattern matching at that stage, with dynamic evidence instead attributed to the execution step. A patch accepted by this project's two-check criterion has only been shown to make one known exploit request fail while a separate health endpoint still succeeds; this is not general correctness, not freedom from all other exploitable inputs, and (established by direct trace inspection in Chapter 4) not even reliably regression-safe, since the criterion cannot distinguish a deliberate security fix from an unrelated crash bug in the generated patch, both of which produce the same observable pattern (the exploit request fails, the separate `/health` endpoint still returns 200). Neither check currently confirms that the patched route itself still behaves correctly for a legitimate, non-malicious request (only that the application process as a whole remains alive) which is a concrete direction for strengthening the validation criterion, not merely a hypothetical one, since Chapter 4 reports a case where this exact gap produced a false acceptance. A corrected validator, introduced later in Chapter 4 (Section 4.1.2), strengthens the exploit side of exactly this gap by distinguishing a deliberate rejection from a crash; its own benign-route check, however, was itself found to require only crash-absence rather than correct behaviour, so this limitation is narrowed, not resolved, and the same caution applies to reading its own results. The free5GC reachability case study's four handlers were originally verified only to compile against the real upstream dependency graph, which is a real but bounded result: it does not by itself constitute dynamic confirmation, and one handler is additionally known to be only partially correct under the project's own rule-based patcher (Chapter 4), fully preserving compilation but not fully closing its handler's vulnerability under that generated patch (the real upstream fix used for runtime testing does not share this limitation). All four handlers were subsequently taken further, by a narrower route than a full core-network deployment: neither the minimal MongoDB-only harness nor its fuller successor stands up free5GC's other network functions (AMF, SMF,

and the rest), and the fuller deployment's real NRF was run with OAuth2 enforcement disabled, so this remains real dynamic confirmation for this one file's four handlers under that specific configuration, not evidence that free5GC's broader core-network behaviour, or this same vulnerability under OAuth2 enforcement, would confirm the same way under this stronger standard. Finally, LLM-dependent pipeline stages were first verified using stubbed model responses to establish that the surrounding logic functioned correctly, and were subsequently confirmed to operate under live model generation for exploit-artifact generation, self-improvement, and patch generation alike: every patch attempt in the eight-model local catalogue comparison (Section 4.1.5) used real, metered Ollama calls, not a stub; the later 36-run hosted-Claude comparison (Section 4.1.6) and the further OpenAI/Codex/AIxTech sweeps (Section 5.5.1) likewise used real, metered calls to their respective backends, not Ollama. The reachability engine's own separate LLM-based patch fallback was initially verified only via stub, since the free5GC case study's original patches were produced by its rule-based path instead; this was subsequently closed (Section B.4) by asking real local and hosted models to generate the fix for the one handler the rule-based patcher under-fixes, from scratch, with 6 of 8 local models (7 compiled; one, `qwen2.5-coder:1.5b`, compiled but was only partially correct under runtime testing) and 7 of 8 hosted Claude models producing a compile- and runtime-confirmed correct patch.

Chapter 4

Results and Discussion

4.1 Results

4.1.1 Exploit Confirmation Across the CVE Catalogue

The five rounds reported in this section are five evolving pipeline runs, not repeated trials of one constant, held-fixed system: code, prompts, the persisted-lesson store, and (for round 5) sampling parameters all changed between at least some rounds, as detailed alongside each round's own description below. They should not be read as a controlled model comparison in the way Section 4.1.5's single-protocol, single-snapshot eight-model comparison can be. Table 4.1 shows the number of catalogued CVEs dynamically confirmed exploitable across five successive verification rounds, each run against the same six CVEs (one per vulnerability class) under a fixed, single-shot-per-CVE protocol within that round. Throughout this chapter, a "confirmed" flag means the described bug pattern was reproduced and dynamically triggered in a generated stand-in target (Section 3.10); it is not independent evidence that the corresponding named upstream package is exploitable when actually installed and run.

Table 4.1: Dynamic confirmation rate by round.

Round	**Confirmed**	**Change from previous round**
Round 1	1/6 (17%)	Baseline (first round)
Round 2	2/6 (33%)	net +1: gained SSRF and XSS, lost SQLi (confirmed round 1)
Round 3	1/6 (17%)	−1 (net; Path Traversal newly confirmed, two prior confirmations lost)
Round 4	4/6 (67%)	+3
Round 5	5/6 (83%)	+1

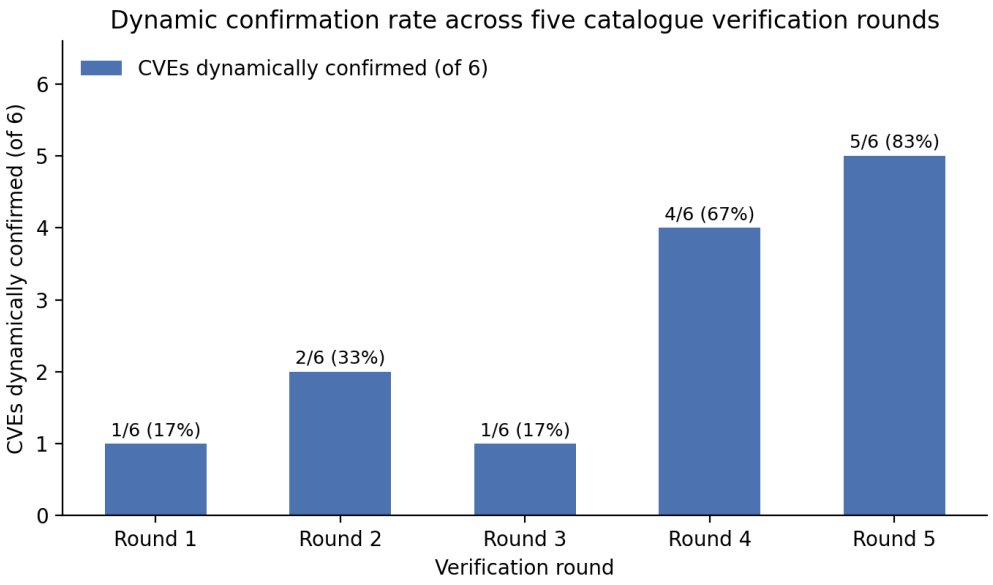


Figure 4.1: Dynamic confirmation rate across the five catalogue verification rounds (Table 4.1). The non-monotonic dip at Round 3 and the jump at Round 4 are discussed in Section 4.2.1.

Considered across all five rounds rather than any single round in isolation, all six vulnerability classes in the catalogue confirmed dynamically at least once; no class was permanently unconfirmable. Round 4 corrected two implementation defects that had been suppressing round 3's result: an unstripped markdown code fence in model-generated output, present in the execution log of four of round 3's five unconfirmed cases, and a stale exit-code/confirmation field that was not reset when a target crashed before its health check. Zero of round 4's six reports and zero of round 5's six reports contained the fence-crash signature, checked programmatically against every execution log.

4.1.2 Patch Generation and Validation

Across rounds 1–3 of the main pipeline's live-LLM catalogue runs, zero of the attempted patches were accepted by both required conditions of the two-check criterion (the original exploit failing against the patched target, and the target's health check continuing to pass). Round 4 and round 5 each produced one patch accepted by this criterion, in both cases for CVE-2026-78683 (Insecure Deserialization) and in both cases resulting in the patched target returning HTTP 500 when the original exploit was re-run against it, satisfying the pipeline's first condition.

Trace-level inspection of both patches found that the two HTTP 500 responses have different, and differently trustworthy, causes. Round 4's patched handler deliberately raises `ValueError("Deserialization only allowed for safe classes")` from an added class whitelist: consistent with the patch summary's own description and plausibly

a genuine security fix, though this project's evidence for it stops at that trace-level reading, not an independent oracle confirming the fix is correct. Round 5's patched handler, by contrast, raises `NameError: name 'io' is not defined`: the generated patch called `io.BytesIO(...)` without importing the `io` module, a plain implementation bug in the generated code, not a security check of any kind. Both traces are reproduced in full in Appendix B. Because the pipeline's two-check acceptance criterion treats *any* exception on the vulnerable route, deliberate or accidental, identically to a successful block, it could not and did not distinguish these two cases: round 5's `patch_validated: true` flag is a **false acceptance** by trace-level inspection, not a verified repair, and this project's own automated pipeline cannot currently tell the two apart on its own. This is a genuine, structural limitation of the Stage 3.8 acceptance criterion's design.

Round 5's patch was genuinely re-probed with its two recorded alternate payloads: the run's raw terminal log records two separate `Multi-payload re-probe` calls, each returning "still blocked." This does not, however, establish a robust fix. The JSON report preserves only the original payload's full trace, not an individual trace per alternate payload, so the specific mechanism behind each "still blocked" result cannot be independently confirmed from the report alone; and the preserved trace shows the `NameError` occurs in a line that constructs the unpickler before any payload-specific branching is reached, which is consistent with the crash being independent of payload content; this reading is a logical inference from the traceback's location in the code, not something separately confirmed by a preserved benign-request trace or the generated patched source itself, which is not preserved outside its temporary sandbox path (Appendix D). The two alternate-payload probes therefore ran, but at minimum re-confirm the same crash rather than demonstrating the underlying vulnerability was closed.

A corrected validator

The false-acceptance finding above was addressed directly, not merely documented. The two-check criterion was extended, additively, with two new checks and a headline verdict field on the same underlying report object: `patch_exploit_blocked` (did the exploit fail *without* an unhandled exception in the target's own process log, checked via a crash-signature pattern over that log rather than a separate instrumentation channel), `patch_function_preserved` (did a fixed, non-malicious benign request to the same route complete *without* that same crash-signature pattern appearing: precisely what this field does and does not establish is stated below, since it is weaker than it may first appear), and a headline `patch_verdict` taking one of four outcomes: `confirmed_fix` (blocked and preserved), `regression_broke_route` (blocked but the benign request also now fails), `inconclusive_crash` (the exploit request itself crashed the route, so blocking cannot be attributed to a deliberate check), and `not_blocked` (the exploit still succeeds).

The original `patch_validated` flag is left completely unchanged so every prior round's reported number stays under its original criterion; the new fields are read only where explicitly cited. Replaying the round 5 false-acceptance trace from Appendix B through the corrected validator (with execution mocked to reproduce the exact preserved log) reproduces `patch_validated: true` as before, while the new `patch_verdict` field correctly reads `inconclusive_crash`: confirming the corrected criterion catches, automatically and without manual trace inspection, exactly the case that previously required it.

What `patch_function_preserved` does and does not establish. The benign probe reuses the same generated proof-of-concept script with a fixed, generic non-malicious payload substituted for the exploit's own payload, rather than a class-specific request with a known-correct expected response. `patch_function_preserved` is set to true when that probe's target process executes and the same crash-signature pattern used for `patch_exploit_blocked` does not appear in its log: it does *not* require the benign request to return HTTP 200, does not require the reused proof-of-concept script to itself report success, and does not check any class-specific expected output. A direct audit of every `confirmed_fix` verdict in the 36-run hosted-Claude comparison (Section 4.1.6), prompted by external review, found that all 27 such verdicts' benign probes were reported as failed by the reused proof-of-concept script's own logic (since a generic payload does not trigger that script's exploit-specific success marker), and that only 13 of those 27 returned HTTP 200 on the benign request's final response, with the remaining 14 returning 401, 403, or 400: a clean HTTP-level rejection, not a crash, but also not confirmation that the route handled the legitimate request as intended. A closely related, narrower case in the same dataset: one `confirmed_fix` verdict (Haiku 4.5, CVE-2026-42208) followed an exploit re-probe whose own log read `[-] Target unreachable` rather than a clean HTTP rejection or a crash signature; a separate, later benign probe against the same patched target started and responded normally, indicating the "unreachable" reading was most likely a transient startup race in the reused script's own health check rather than a deliberate block, which the crash-signature check (built to separate a deliberate rejection from a crash) does not distinguish from either. This is a genuine limitation of the corrected validator, not a claim it does not have, and every reading of `patch_verdict` = `confirmed_fix` in this project from Section 4.1.6 onward should be understood as "the exploit was rejected without a detected crash, and a generic benign request to the same route also completed without a detected crash," not as independent confirmation that the route's legitimate behaviour is correct.

A stronger, content-verified reassessment. This limitation was addressed directly, not left as a stated gap: for each of the six main-catalogue vulnerability classes, a real expected malicious result and a real expected legitimate result (specific HTTP status and exact response content, read from the actual generated target/proof-of-concept source rather than assumed) were defined and re-tested against surviving patched artifacts from the 36-run comparison. Locating those artifacts surfaced a further, previously

undocumented harness gap: Stage 3.5/3.8's artifact writer saves generated files to a shared, non-model-specific path regardless of a run's own output destination, so every model that touched the same CVE in the comparison overwrote the same file: only 6 of the original 36 patched targets survive on disk, one per CVE (whichever model happened to run last for it), not all 36. Re-testing those 6 survivors under the stronger, content-verified criterion, every one passed: each malicious request received a clean, class-appropriate rejection distinct from the vulnerable version's response, and each legitimate request received its exact expected content, not merely a crash-free one. This reassessment also explained an apparent contradiction found while building it: `claude-opus-4-8`'s surviving CVE-2026-46492 patch was originally recorded as `not_blocked`, yet directly tested here as genuinely correct against the proof-of-concept's default payload; the original multi-payload bypass check's alternate payload was found, on inspection, to trigger the generated target's own vulnerability-signalling self-check (a keyword-pattern regex over already-escaped, inert output) without the underlying markup ever being capable of executing in a real browser: a fidelity limitation of the generated target's own signal, not a real bypass of the patch. Full methodology, per-CVE expected values, and this finding: `reports/patch_reassessment_2026-09-26/RESULTS.md`. This reassessment does not extend to the other 30 of 36 original attempts, whose patched source no longer exists and cannot be reassessed without new generation cost, and the 6 survivors are not a representative sample of the full comparison.

Table 4.2: The six surviving patched artifacts under the stronger, content-verified reassessment: expected vs. actual response on both the malicious and legitimate request.

CVE (class)	**Expected malicious outcome**	**Expected legitimate outcome**	**Malicious**	**Legit**	**Verdict**
CVE-2026-42208 (SQLi)	No injection marker	Authenticated JSON response	401	200	`confirmed_fix`
CVE-2026-27602 (OS CmdI)	No command-output marker	"Domain created" JSON response	400	200	`confirmed_fix`
CVE-2026-23949 (Path Trav.)	No traversed-file content	Example-file JSON response	400	200	`confirmed_fix`
CVE-2026-78683 (Deserial.)	No RCE marker	"Loaded" JSON response	400	200	`confirmed_fix`
CVE-2026-54729 (SSRF)	No internal metadata leaked	External fetch succeeds normally	403	200	`confirmed_fix`
CVE-2026-46492 (XSS)	No unescaped script markup	Exact escaped heading/paragraph	200	200	`confirmed_fix`

All 6 surviving patches pass this stronger, content-verified test: every malicious request receives a clean, class-appropriate rejection distinct from the vulnerable version's response (never the same HTTP 200-with-marker response the vulnerable version returns), and every legitimate request receives its exact expected content, not merely a crash-free response. This is the full detail behind the single-sentence summary above, reproduced here because the artifacts themselves are not retained in the repository outside their original, now-overwritten scratch paths (Appendix D); the source of record for these six rows is `reports/patch_reassessment_2026-09-26/RESULTS.md`, not a path to the (no longer surviving) individual artifacts.

A second, genuine implementation gap was found while first exercising the corrected validator against live, varied data rather than only the two originally planned synthetic test cases: the code path taken when a patched target's health check fails *entirely* (the target process never becomes reachable at all, the clearest possible crash signature) returned before reaching the new verdict-computation block, leaving `patch_verdict` blank instead of classified. This was fixed by explicitly setting `patch_verdict = inconclusive_crash` in that specific early-return branch, and verified by re-running the affected case and

confirming the corrected output. This is reported here as a further instance of the same pattern already established in Sections 4.2.1 and 4.2.2: a validation harness's own correctness is not established merely by writing it, only by exercising it against real, varied failure modes and checking the result by hand at least once.

The corrected validator was then applied to a sixth full-catalogue verification round, run under the same isolated-lessons-store protocol as rounds 1–5 (Appendix A) and intended, in its original design, to compare the Ollama backend against a hosted Claude Code CLI session as a second live-model backend. That specific comparison did not succeed: the `claude` CLI's authentication session had silently expired between the CLI's own version check (which does not require authentication and reported success) and its actual generation calls (which failed with an authentication error on every attempt), and the pipeline's existing backend-selection logic treated this as an ordinary unavailable-backend condition and silently substituted the local Ollama backend for every call, with no warning surfaced anywhere in the run's own logs. This was caught only by manually auditing the `generation_backend` field recorded on each report against the run's own stated protocol, after the run had already completed and been reported once as a Claude Code comparison. The pipeline's silent-fallback behaviour was subsequently hardened so that a caller requesting a specific model explicitly is no longer silently substituted without a visible warning; the round's real, honest-but-mislabelled data was kept and reported as a sixth Ollama round rather than discarded, once relabelled correctly. Under the corrected validator, this round's four confirmed exploits each received a patch attempt, and every one of those four patches failed to close the vulnerability under the new criterion: three as a clean `not_blocked` and one as `inconclusive_crash`: the same qualitative pattern as round 5, now distinguished explicitly rather than collapsed into a single flag, and zero `confirmed_fix` results in this specific round.

A separate, dedicated investigation was carried out specifically to test whether a patch accepted by the two-check criterion could still be bypassed by a different payload for the same vulnerability class, rather than trusting a single-payload pass as sufficient. This investigation surfaced and fixed four further implementation defects in the patch-generation and acceptance path, distinct from the exploit-generation defects described in Section 4.1.1: an unstripped markdown code fence in the model's patched-file response, causing an immediate syntax error; a patched file that omitted Flask's required `app.run()` startup block, causing a silent, traceback-free crash; a diagnostic log that discarded the patched process's real standard-error output, making the first two defects undiagnosable until fixed; and a `localhost` hostname that resolved to two different locally bound services (IPv4 and IPv6) on the development machine, causing model-availability checks to return inconsistent answers from call to call. After fixing all four, a full eight-model, six-CVE re-test found 11/48 exploits confirmed, 9/48 patches attempted, and 3/48 accepted by the two-check criterion: a substantial rise from the 0/48 accepted in the original, pre-fix

comparison (Section 4.1.3).

This 3/48 figure requires one further, essential qualification before it is reported: all three nominally accepted patches had an empty `payload_variants` field, meaning the multi-payload bypass check (built specifically to catch a patch that blocks only the one payload tried) had no alternate payload to probe them with, because the small model that produced them (`qwen2.5-coder:1.5b`) did not successfully generate the alternate-payload section in those cases. A controlled synthetic test independently confirmed the bypass check itself works correctly: a deliberately narrow patch that blocklisted only the literal default payload passed the single-payload check but was correctly rejected once a structurally different payload was tried. The honest, precise headline for this specific 48-run retest is therefore not "3/48 patches validated," but **0/48 patches in this retest have been validated against more than their own original payload**: the bypass-testing mechanism is built and proven correct, but no patch in this particular dataset was exercised against it with a real alternate payload. This is narrower than a claim about the whole project: the main catalogue's own round-5 patch genuinely was re-probed with two alternate payloads, though that probe turned out to be uninformative for a different reason (Section 4.1.2).

4.1.3 Real-Upstream Validation: CVE-2026-23949

Every result reported so far in this chapter, aside from the free5GC case study, exploits a generated Flask reproduction of an advisory's described bug pattern, not the actual named upstream package (Section 3.10). One case was carried out instead against the real, pinned-version package, to close part of that stated gap directly rather than leave it purely as a limitation. CVE-2026-23949 (`jaraco.context`, a path-traversal defect in its `tarball()` helper) was selected as a tractable candidate: the advisory names a specific vulnerable range and a specific fixed version, and the vulnerable function's real behaviour could be exercised without any special deployment beyond a Python virtual environment.

Two isolated virtual environments were built, one installing the real, unmodified `jaraco.context==5.3.0` (inside the advisory's own stated vulnerable range) from PyPI, the other installing the real, unmodified `jaraco.context==6.1.0` (the advisory's own stated fixed version). The installed source in both environments was read directly to confirm the mechanism rather than assumed from the advisory text alone: `5.3.0`'s `strip_first_component` filter splits a tar member's path on its first `/` and keeps everything after it, including any `../` sequences in the remainder, while `6.1.0`'s filter composes Python's own standard-library `tarfile.data_filter` (PEP 706) ahead of the same function. A real malicious tar archive (a legitimate-looking entry alongside one crafted to escape the extraction directory by two levels) and a real, non-malicious tar archive were built and served from a local, self-controlled HTTP server, exactly matching the

local-target discipline used throughout the rest of this project's dynamic evaluation; the real, unmodified `jaraco.context.tarball()` function was then called directly from each environment's own interpreter, with no mocking of the library under test.

Table 4.3 reports the full four-way result. Read against the vocabulary introduced in Section 4.1.2: `patch_exploit_blocked` is true on `6.1.0` because the block is a named, documented standard-library exception (`tarfile.OutsideDestinationError`) raised deliberately by PEP 706's own extraction-safety filter, not an unrelated crash signature; `patch_function_preserved` is true because the tested legitimate archive's extracted file content was independently read back and confirmed to match the expected content on both versions, a stronger, content-verified check than the generic crash-absence probe used for this same field name in the main-catalogue comparisons (Section 4.1.2): this stronger check was practical here specifically because the legitimate-use behaviour of a tar-extraction function has one clear, checkable expected output, which is not true of every vulnerability class in the main catalogue. The case therefore reaches `patch_verdict = confirmed_fix` on this project's strongest evidence tier among the Python-based main catalogue's cases, since the "patch" here is the real upstream maintainers' own released fix rather than an LLM-generated one, and the benign check is content-verified rather than merely crash-absent; this remains one tested legitimate-archive case, not exhaustive proof the fix preserves every legitimate use.

Table 4.3: Real-upstream four-way validation result: CVE-2026-23949 (`jaraco.context`).

Version	**Test**	**Result**
`5.3.0` (vulnerable)	Malicious archive	Escaped: canary file written outside the extraction directory, exit 0
`5.3.0` (vulnerable)	Legitimate archive	Extracts correctly
`6.1.0` (patched)	Malicious archive	Blocked: `tarfile.OutsideDestinationError`, exit 1
`6.1.0` (patched)	Legitimate archive	Extracts identically to the vulnerable version

This result does not generalise to the other five main-catalogue CVEs, whose targets remain generated reproductions, and it does not involve any LLM-generated patch: it validates this project's own *evaluation methodology* (that a real exploit can be demonstrated and a real fix confirmed against real, named, installed software using this project's own evidence-gating discipline) independently of whether an LLM can produce a patch this good, which remains the separate, harder question this chapter's other results address.

4.1.4 Real-Upstream Validation: OpenEMR (GHSA-q366-cv5v-83w8)

A third real-upstream case transferred this project's evidence-gating method to a second application domain, healthcare, rather than confirming it again within the same telecom/5G or Python-packaging contexts already covered by free5GC and `jaraco.context`. GHSA-q366-cv5v-83w8 is an unauthenticated information-disclosure advisory in OpenEMR's `admin.php`, carrying no assigned CVE number (its `cve_id` field is `null`, confirmed via the GitHub API); the real, unmodified OpenEMR source was built and run at its vulnerable commit (`b5313ea`) and at the real upstream fix commit (`50f789f`, PR #13133), with no reproduction or stand-in target. The fix is structurally the same class as free5GC's missing-`return` defect: an early-exit opt-in guard added before any other logic in the file, returning `403` unless `OPENEMR_ADMIN_PHP_ENABLED` is explicitly set, rather than a rewrite of the underlying disclosure logic.

The real, unmodified OpenEMR checkout was bind-mounted into a minimal Docker deployment (a real, official `mariadb:10.11` image seeded with only the two tables `admin.php`'s read path actually needs, and a minimal `php:8.2-cli` image running PHP's own built-in development server, no Apache and no full OpenEMR setup wizard) the same minimal-dependency-scoping principle already applied to both free5GC harnesses (Section 3.8.2). Table 4.4 reports the full three-request result.

Table 4.4: Real-upstream validation result: GHSA-q366-cv5v-83w8 (OpenEMR `admin.php`).

Request	**Expected**	**Actual (this run)**
Vulnerable commit, no env var	`200`, metadata disclosed	`200`: rendered table shows Site ID, DB name, Site Name, and version pulled directly from the seeded database rows
Patched commit, no env var	`403`, metadata blocked	`403`, body exactly `admin.php is disabled by default...`: no site, database, or version data anywhere in the response
Patched commit, `OPENEMR_ADMIN_PHP_ENABLED=1`	`200`, admin page works	`200`, content identical to the vulnerable run's disclosure: confirms the fix is a default-closed gate, not a removal of the underlying function

Read against the vocabulary in Section 4.1.2: the patched build's legitimate-use request

was checked against its exact expected content (the same disclosure fields the vulnerable build showed), not merely crash-absence, giving this case `patch_verdict = confirmed_fix` on this project's strongest, content-verified evidence tier: the real upstream maintainers' own released fix, exercised against the real, unmodified application rather than a reproduction or an LLM-generated patch, matching `jaraco.context`'s standard of evidence rather than the heuristic crash-absence screening used for the main catalogue's LLM-generated patches. Running this harness for real surfaced two further genuine implementation defects, consistent with this project's established pattern (Sections 4.2.1, 4.2.2, and 3.8.2): the official MariaDB image's entrypoint can report an internal, password-less bootstrap instance ready over a ping check moments before the real, fully-initialised server with the committed root password actually accepts logins, and PHP's built-in development server can accept a TCP connection fractionally before its request-handling loop is serving requests, resetting the very next request with a bare connection close. Both were fixed by replacing the naive readiness check with the actual evidence-gathering action (an authenticated schema-load login; a real HTTP request retried briefly on connection error) rather than a weaker proxy for it.

This case does not generalise beyond its own one file and one safe, unauthenticated oracle: it is not a systematic security assessment of OpenEMR, and the minimal two-table schema demonstrates this specific vulnerability's logic precisely without demonstrating anything about OpenEMR's broader application behaviour or authentication system. It was subsequently also used, alongside free5GC, as a target for the later LLM patch-generation and adversarial bypass-probe comparisons across the Claude CLI, AIxTech gateway, and OpenAI API backends reported in Section 4.1.6 and the final evidence freeze (Section 5.5.1); those later comparisons evaluate LLM-generated patches against this same real OpenEMR target, not this section's own real-upstream maintainer fix.

4.1.5 Multi-Model Comparison

Table 4.5 summarises results from running the identical, sampling-pinned protocol (temperature = 0, fixed seed) across eight local Ollama-served models, 48 runs in total.

Table 4.5: Confirmation rate by model (6 CVEs each).

Model	Confirmed	Patch accepted	Total LLM call time (s)
qwen2.5-coder:1.5b	3/6*	0/6	37.3
qwen2.5-coder:3b	1/6	0/6	38.4
qwen2.5-coder:7b	5/6	0/6	114.9
llama3.1:8b	2/6	0/6	94.3
mistral:7b	0/6	0/6	106.8
gemma2:9b	0/6	0/6	763.7
codellama:13b	2/6	0/6	1544.1
deepseek-coder-v2:16b	4/6	0/6	176.6

*All three of qwen2.5-coder:1.5b's confirmations came from the pipeline's template fallback path, not from the model's own live-generated output: every one of its six generation attempts was recorded as `llm_live_failed`.

Patch acceptance was 0/48 across every model in this comparison. This specific comparison run predates four patch-acceptance harness bugs found and fixed in a dedicated follow-up investigation, detailed in Section 4.1.2; a later, separate 48-run re-test with those fixes applied found a different confirmed-exploit count (11/48, not 17/48) and 3/48 nominally accepted patches, with the important caveat, also detailed in Section 4.1.2, that trace-level inspection of nominally accepted patches from the same round elsewhere in this project found at least one to be a false acceptance rather than a genuine fix. LLM call time did not scale monotonically with parameter count: codellama:13b's total call time (1544.1s) exceeded deepseek-coder-v2:16b's (176.6s) despite the latter having more parameters, and gemma2:9b recorded one single-CVE call time of 608.7s against a 24.8–40.5s range for its other five runs, with no correspondingly large output-token count. This column measures model generation time specifically, not each run's full pipeline wall-clock, which also includes subprocess execution, health checks, and I/O outside the model call itself.

4.1.6 Multi-Model Comparison: Hosted Claude Models

The eight-model comparison above evaluates only locally hosted, open-weight models; no hosted API or CLI session was available in this project's environment at the time. A second, later comparison became possible once a Claude Code CLI session was authenticated in this environment, and was carried out across six named Claude model snapshots (Haiku 4.5, Sonnet 4.6, Sonnet 5, and Opus 4.6/4.7/4.8) against the same six-CVE catalogue, under the same one-shot-per-CVE, isolated-lessons-store protocol as every Ollama round, with the corrected validator's `patch_verdict` (Section 4.1.2) as its headline

patch metric from the outset: read throughout this subsection as heuristic patch-screening (exploit rejected without a detected crash, and a generic benign probe also completing without a detected crash), not as confirmed functional correctness, per the precise scope stated in Section 4.1.2.

Table 4.6: Confirmation and patch outcomes by hosted Claude model (6 CVEs each, 36 runs total).

Model	**Confirmed**	**Screening pass (`confirmed_fix`)**	**Not blocked**	**Crash**	**Cost (USD)**
Haiku 4.5	6/6	5	1	0	$1.81
Sonnet 4.6	6/6	4	2	0	$4.50
Sonnet 5	6/6	4	2	0	$4.26
Opus 4.6	6/6	4	1	1	$8.04
Opus 4.7	6/6	5	1	0	$5.52
Opus 4.8	6/6	5	1	0	$9.04
Total (36 runs)					**$33.18**[*]

[*]Computed from each model's unrounded per-call cost, not from summing the rounded cells shown above (which sum to $33.17); the one-cent difference is a rounding artifact, not a data error.

Note on the Screening pass column: as throughout this chapter, `confirmed_fix` here means the two-check criterion passed (original exploit blocked, benign probe completed without a detected crash), not that the patch was independently verified correct. Only 6 of the 36 underlying patched artifacts from this comparison survive for stronger, content-verified re-testing (Section 4.1.2 explains why); the counts in this column should be read as heuristic screening results at this scale, not as 27 independently confirmed correct patches.

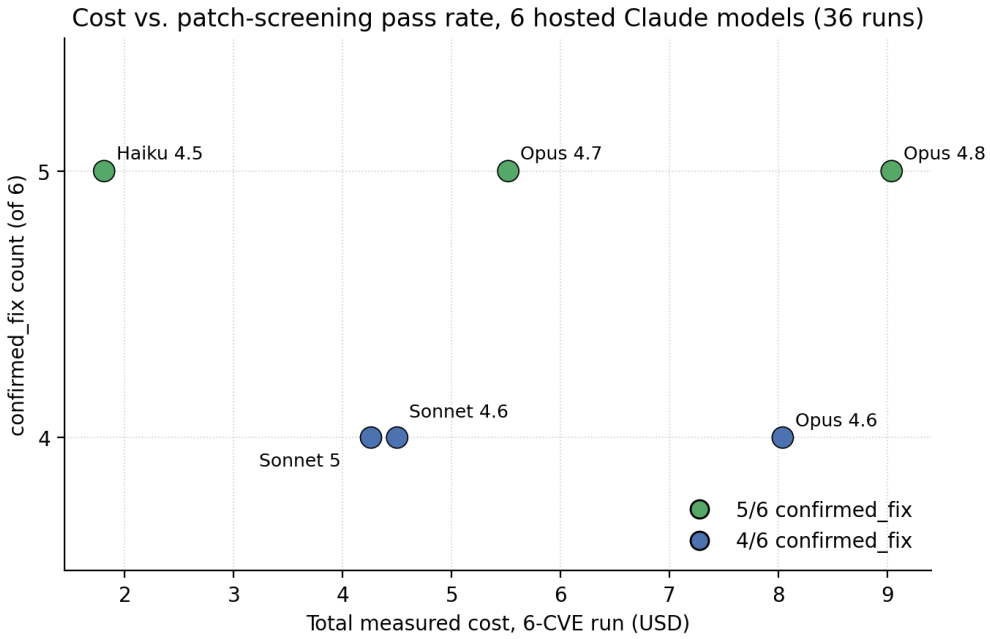


Figure 4.2: Total measured cost against `confirmed_fix` count for each of the 6 hosted Claude models (Table 4.6). Cost and screening-pass rate are visibly not monotonically related: Haiku 4.5, the cheapest model, ties the best rate.

Every one of the 36 runs dynamically confirmed its assigned exploit, so (unlike the eight-model local comparison, where confirmation rate itself varied substantially by model) exploit generation is not the differentiator among these six hosted models at this task's scale; passing the heuristic patch-screening criterion is. Two findings stand out beyond the raw `confirmed_fix` counts in Table 4.6, both read under the scope stated in Section 4.1.2 rather than as confirmed patch correctness. First, cost and screening-pass rate are not monotonically related: Haiku 4.5, the cheapest model evaluated, matched the best `confirmed_fix` rate (5/6) at roughly one third the total real cost of every other model compared, while Opus 4.6, a mid-tier-cost model, produced the single worst outcome in the entire matrix: its one miss is `inconclusive_crash`, the only crash-class failure across all 36 runs, despite Opus 4.6 being neither the cheapest nor the fastest model tested. Second, one vulnerability class (Cross-Site Scripting (CVE-2026-46492)) was the sole class every one of the six models failed to pass even this heuristic screen (`not_blocked` in every case), a consistent, cross-model finding rather than one model's individual weakness, suggesting whatever makes this specific class's patch harder to generate is a property of the task or the prompt, not of any one model in this comparison.

Table 4.6 reports only per-model totals across all six CVEs; Table A.5 in Appendix A gives the full per-CVE breakdown (individual cost and wall-clock time for every one of the 36 model–CVE cells) since a per-model total alone hides real variation across CVEs within the same model (for example, Opus 4.8's cost per CVE ranges from $0.81 to $2.13,

and Sonnet 4.6's wall-clock time ranges from 80.3s to 257.8s, both within a single model's own six runs).

This comparison surfaced its own harness defects, in keeping with this project's established pattern of validation issues surfacing only once a harness is actually exercised against real, varied data (Section 4.1.2). The Claude CLI backend had never previously exposed real per-call cost or token counts: its plain-text output mode returns no usage metadata at all, so every prior Claude-CLI call in this project's history recorded `cost_usd: None`. Switching to the CLI's structured JSON output mode exposes a real `total_cost_usd` and token breakdown directly from the provider, which is what makes Table 4.6's cost column possible at all. Two further defects were found and fixed while running this specific comparison: the same silent-fallback behaviour that had already affected the round-6 catalogue run (Section 4.1.2) recurred under a genuine account rate-limit condition mid-comparison, hardened at the source rather than merely documented a second time; and two of the six models (Opus 4.7 and Opus 4.8) initially failed the majority of their assigned generation calls because they sometimes attempted to invoke an interactive file-write tool rather than returning plain text, which stalls indefinitely in this pipeline's non-interactive calling context: resolved by explicitly disabling tool use and cross-call session persistence for every Claude CLI invocation, after which both models completed their full six-CVE run cleanly. As with the eight-model local comparison, this is a single deterministic-in-intent but not sampling-pinned shot per CVE per model (the Claude CLI exposes no temperature or seed control), and should be read as one data point per model rather than a statistically powered ranking.

4.1.7 Reachability Filtering on Real Upstream Source

Applied to a real, 86 KB free5GC/UDR source file, the reachability engine parsed 102 functions and identified 4 as reachable from four HTTP entry-point handlers selected manually as the module's externally callable surface, a 96.1% reduction in the function count requiring further analysis. All four reachable functions were flagged by the rule-based pattern matcher and classified as CWE-285 (improper authorisation check), matching the real, disclosed bug class; this classification was checked against the advisory's own CWE label, not confirmed by an independent second reviewer. A patch was generated for all four; applying all four patches to a live clone of the actual `free5gc/udr` repository at the pre-fix commit, `go build ./...` completed successfully against the module's genuine dependency graph, both before and after the patches were applied. One of the four generated patches is only partially correct: its target handler (`HandleApplicationDataInfluenceDataSubsToNotifyGet`) requires two separate `return` insertions in the real, disclosed fix, while the rule-based patcher inserts only the first match per function body, so that handler's vulnerability is not fully closed even though the module still compiles. This partial-correction limitation is

specific to this project's own rule-based auto-patcher, not to the real upstream fix used for the runtime testing described next: all four handlers were subsequently taken beyond this original compile-time verification to real runtime confirmation, as described below. This specific failure mode (a patch that addresses the fix site a developer or patcher first notices while leaving a structurally identical second site unaddressed) is not an idiosyncrasy of this project's rule-based patcher: a large-scale study of incomplete security fixes in the Linux kernel identifies "overlooked similar modules" as one of three recurring root causes of incomplete patches found across its dataset, alongside a developer being misled by the crash's superficial symptoms rather than its underlying cause [20].

A fourth handler, the DELETE subscription endpoint (`HandleApplicationDataInfluenceDataSubsToNotifySubscriptionIdDelete`), sharing the same missing-`return` root cause as the other three handlers tracked by this project under CVE-2026-40248's real fix commit (the public advisory's own text describes unauthenticated creation and modification specifically, not this project's own extension to the sibling DELETE/GET behaviour; see Appendix A for the full per-CVE evidence table and how CVE-2026-40248 and the separately-evaluated CVE-2026-40246 are kept distinct), was subsequently taken one evidence tier further using the harness described in Section 3.8.2. The real, cloned upstream module was built and run twice (once at the vulnerable commit, once at the real fix commit) against a real local MongoDB instance, and driven with real HTTP requests rather than compiled but never executed. At the vulnerable commit, the malicious delete request (using an invalid `influenceId` path segment) received a `404` response, matching the handler's own validation check, but a follow-up read confirmed the targeted subscription record had been deleted regardless: the vulnerable code writes the `404` response but does not `return` afterward, so execution falls through and performs the deletion anyway. At the real fix commit, the identical malicious request again received `404`, but the follow-up read confirmed the record was still present: the added `return` statement stops execution before the deletion path is reached. A legitimate delete request, using the correct `influenceId` value, succeeded and removed the record on both builds, confirming the fix does not break the handler's intended function. This gives a real, non-inconclusive verdict under the pipeline's own `patch_verdict` field (`confirmed_fix`) for this one handler, obtained by direct runtime evidence rather than compilation alone. This result upgrades the free5GC case from compile-time evidence to runtime evidence. The vulnerable UDR commit was shown to execute the unintended deletion despite returning a 404 response to the client, while the patched commit returned before the deletion path and preserved the record. The benign deletion path remained functional, giving this case the project's strongest patch-validation evidence among its compiled, real-upstream reachability work. The full evidence bundle (process logs, HTTP request/response logs, the build logs for both commits, the generated `manifest.json` and `verdict.json`, and the real upstream diff between the two commits) is described in

Appendix A. At the time this minimal harness was run, this remained a single-handler result, not evidence that the other three reachable handlers would behave identically under the same runtime standard; that gap was subsequently closed, as described next.

All three remaining handlers were then run against the fuller Docker-based deployment described in Section 3.8.2 (real MongoDB, real free5GC NRF, OAuth2 disabled), and were found to share the DELETE handler's exact root cause (the same missing `return` after a non-2xx write) with two of the three found to be more severe than the DELETE case on direct empirical inspection of the raw HTTP response bytes. The single-record `GET` handler, queried with an invalid `influenceId`, returns `404 Not Found` with the literal body `"404 page not found"` on the patched build, but on the vulnerable build the same literal text is immediately followed, in the same response, by the real subscription's JSON content: Go's `net/http` frames each handler `Write()` call as its own chunk when no explicit `Content-Length` is set, so the vulnerable handler's fall-through second write is not silently dropped, it is delivered to the client. The collection-level `GET` handler (queried with no filter parameters at all, its own documented error condition) shows the same pattern at larger scale: the vulnerable build's `400 Bad Request` error body is followed by a JSON array of *every* subscription record in the system, since an empty filter matches every stored record: a full, unauthenticated collection dump inside what looks like a rejected request. The single-record `PUT` handler goes further still: the request again returns `404 Not Found`, but a follow-up read confirms the submitted data was genuinely written to the in-memory store: an unauthorized *write*, not only a read leak. All three close cleanly on the patched build (the response body contains only the expected error text, and the follow-up read for the `PUT` case confirms nothing was stored), with the legitimate, correct-`influenceId` request path unaffected on both builds. Combined with the DELETE result above, this constitutes a full CRUD bypass (unauthorized read of one record, unauthorized read of the entire collection, unauthorized write, and unauthorized delete) through one shared, one-line root cause, confirmed and closed across all four reachable handlers rather than one. This work is recorded on a separate branch (`free5gc-full-deployment`, commit `e59ee4e`) with its own evidence bundle (`free5gc_full_deployment/evidence/`), kept isolated from the single-handler result's own evidence bundle so the two can be distinguished; both are cited in Appendix A.

4.1.8 Classification Ablation

A separate ablation isolated Stage 2 (vulnerability classification) from the generation stages evaluated above, testing three conditions against the six-CVE catalogue's ground-truth CWE labels: full advisory content (prose description plus structured CWE field), prose description only, and the CVE identifier alone with no advisory content. Table 4.7 reports class-level match (does the predicted vulnerability class match ground truth) and

exact-CWE match (does the predicted CWE code match exactly) as separate metrics, since collapsing them into a single accuracy figure would obscure a real, informative difference between the two.

Table 4.7: Stage 2 classification ablation (n=6).

Condition	**Class-level match**	**Exact-CWE match**
Prose description only	5/6	4/6
Full advisory (structured CWE field)	6/6	6/6
CVE identifier only (no content)	0/6 (no classification)	0/6 (no classification)

Prose-only classification correctly identified the vulnerability class for 5 of 6 CVEs, with one full miss (CVE-2026-42208, SQL Injection): its NVD summary describes the SQL injection's mechanism without ever using the words "SQL" or "injection," so keyword-based prose matching returned no classification at all. Among the 5 classified correctly at the class level, one (CVE-2026-46492) was assigned the wrong exact CWE: prose matching returned the general CWE-79 for any detected cross-site-scripting pattern, while the CVE's actual, analyst-assigned code was the more specific CWE-80. Supplying the structured CWE field already present in the advisory (rather than relying on regex matching against free-text prose) resolved both cases, reaching 6/6 on both metrics. This full-condition result should be read precisely: it demonstrates that the classifier correctly reads a ground-truth field already present in the advisory, which is metadata recovery, not independent classification of unseen content: the genuinely informative result of this ablation is the 5/6 and 4/6 figures under prose-only matching, and the specific, narrower failure mode (family-level mapping losing subtype precision) that the structured-field condition helps isolate, not the 6/6 figure on its own. The CVE-identifier-only condition, which received no advisory content at all, correctly produced no classification for every entry, ruling out identifier-based hardcoding in either classifier.

A cross-check against GitHub's independent advisory database found CWE agreement with NVD's structured field for all six catalogue entries (one entry, CVE-2026-46492, lists an additional CWE alongside the matching one), indicating the structured-CWE signal is consistent across both databases the ingestion stage can consult, though this agreement should not be read as independent replication of the underlying vulnerability analysis, since advisories are commonly shared or ingested between the two databases.

Precision, recall, and F1 for the prose-only condition

Table 4.7 reports raw match counts; Table 4.8 restates the same prose-only results (this ablation's most informative condition, per the discussion above) as micro-averaged precision, recall, and F1, computed by treating a missing classification as a false negative

with no corresponding false positive, and a wrong classification as both a false negative for the true label and a false positive for the predicted one. With $n = 6$ and exactly one instance per class, recall at the class level and at the exact-CWE level is mathematically identical to the already-reported 5/6 and 4/6 accuracy figures; precision and F1 add the information accuracy alone does not capture: that the exact-CWE condition's errors include one genuine misclassification (a wrong label asserted, lowering precision), not only one abstention.

Table 4.8: Prose-only condition: micro-averaged precision, recall, and F1.

Level	**Precision**	**Recall**	**F1**
Class-level (family)	1.000	0.833	0.909
Exact-CWE	0.800	0.667	0.727

For completeness, the other two conditions are degenerate under this framing rather than independently informative: the full-advisory condition (6/6) scores precision = recall = F1 = 1.000, and the CVE-identifier-only condition (0/6) scores recall = 0.000 with precision undefined (no classification was produced for any of the six entries, so the precision denominator is zero): both exactly as Table 4.7's raw counts already imply.

Table 4.9 gives the exact-CWE confusion matrix for the prose-only condition. Four of six true labels are correctly recovered (the diagonal); the two off-diagonal entries are this condition's only errors and are each individually traced to a specific, named cause in the discussion above: CVE-2026-42208's missing SQL-injection keywords produced an abstention (the *No prediction* column), and CVE-2026-46492's family-level CWE-79 mapping, correct at the class level but imprecise for this CVE's analyst-assigned CWE-80, produced a genuine misclassification rather than an abstention.

Table 4.9: Prose-only condition: exact-CWE confusion matrix (n=6, one instance per true class).

True CWE	22	78	79	80	89	502	918	*No pred.*
CWE-22	1	0	0	0	0	0	0	0
CWE-78	0	1	0	0	0	0	0	0
CWE-80	0	0	**1**	0	0	0	0	0
CWE-89	0	0	0	0	0	0	0	**1**
CWE-502	0	0	0	0	0	1	0	0
CWE-918	0	0	0	0	0	0	1	0

Row/column labels are CWE numbers only; see Table 4.8 and the surrounding discussion for the mapping to vulnerability class names (Path Traversal, OS Command Injection, XSS, SQL Injection, Deserialization, SSRF). The true row CWE-80 (XSS, CVE-2026-46492) is predicted as the family-level default CWE-79 because the prose-only classifier's hand-encoded mapping assigns any detected cross-site-scripting pattern to CWE-79 by default, losing the analyst's more specific CWE-80 subtype judgment: a mapping-precision limitation, not a misclassification of the vulnerability class itself.

4.2 Discussion

4.2.1 Interpreting the Confirmation-Rate Progression

The rise from 1/6 in round 1 to 5/6 in round 5 is not evidence of a steadily improving model or pipeline in the way the raw trend line might suggest. Round 3's drop to 1/6, despite a genuine bug fix being introduced in that round (self-improvement's LLM-revision path was wired to the live backend for the first time), shows that a working retry mechanism is not the same thing as a mechanism that reliably finds the correct fix. It is clearer, and more defensible, to attribute most of round 4's jump specifically to the two implementation-defect fixes rather than to any change in model capability: a plausible reading, given that the same model and largely the same prompt were used across rounds 3 and 4, and the round-4 improvement coincides precisely with the fence-crash signature disappearing from every report. This is a correlational reading, not a causal proof: the round-3-to-round-4 comparison in this project's own records cannot isolate the fixes' individual contribution from other differences between the two runs, and it is offered here as the more plausible explanation rather than an established one. What the records do support is a more general caution, reinforced by the patch-validation false-acceptance finding in Section 4.1.2: a harness defect of this kind can distort a reported result in either direction, suppressing a genuine confirmation (as the markdown-fence crash plausibly did in round 3) or manufacturing a false one, so neither a low nor a high raw count in this project's own catalogue should be read as a reliable floor or ceiling on the pipeline's true capability without independent trace verification of the specific cases involved.

The SQL Injection class's behaviour across rounds (confirmed in round 1, unconfirmed

in rounds 2–4, confirmed again in round 5) is consistent with the project's own stated caveat that a single-shot, non-averaged protocol can flip for the same CVE between runs, since prompt content still varies by CVE and neither the model's sampling nor the surrounding pipeline state is guaranteed identical run-to-run even with a fixed seed. This should be read as a property of the evaluation protocol, not as evidence that any one class is inherently harder than another.

4.2.2 Why Patch Validation Lagged Confirmation

The single most important finding on patch acceptance is not a rate but a design flaw, established in Section 4.1.2 by direct trace inspection: the pipeline's two-check acceptance criterion (original exploit fails, health check passes) cannot distinguish a deliberate security fix from an unrelated crash bug in the generated patch, because both produce an HTTP 500 response that satisfies the first check identically. Round 5's "validated" patch for CVE-2026-78683 is a confirmed false acceptance on this basis. This reframes every raw patch-acceptance count reported in this chapter: a nonzero count should be read as "the automated criterion was satisfied," not as "a correct fix was produced," until each individual case has been traced by hand as round 4 and round 5 were here. Round 4's own accepted patch is plausibly a genuine fix on trace-level reading, but this project's evidence for that claim is the same kind of manual inspection that caught round 5's false acceptance, not an independently stronger automated guarantee. This is not an idiosyncrasy of this project's own harness: PVBench's independent, 209-case benchmark found over 40% of patches three separate state-of-the-art automated-repair systems had validated as correct under basic testing failed its stronger, developer-authored test standard [13], and VulnRepairEval similarly found that moving from superficial validation to exploit-based differential testing collapsed the best-performing model's apparent success rate to 5 of 23 cases [12]. Read against that literature, round 5's single false acceptance is a small-sample instance of a pattern now independently measured at benchmark scale, not a defect unique to this project's own two-check criterion.

Two distinct 48-run datasets are reported in this chapter and should not be collapsed into one figure. The initial eight-model comparison (Section 4.1.5) found 17/48 exploits confirmed and 0/48 patches accepted. A later, separate re-test, carried out after four implementation bugs in the patch-validation harness were found and fixed (an unstripped markdown fence, a patch missing Flask's `app.run()` block, a discarded diagnostic log, and a `localhost` resolution ambiguity), found 11/48 exploits confirmed and 3/48 patches nominally accepted: a different run, with a different confirmed-exploit count, not the same 48 records re-scored. Given the false-acceptance finding above, even the corrected 3/48 figure should be read cautiously rather than as a straightforward improvement: the two runs contain different confirmed-exploit cases under an acceptance criterion that

can, as shown for round 5, still accept a patch whose vulnerable route simply crashes. The four bug fixes changed specific, observed behaviours (patches no longer crash on a stray markdown fence or a missing `app.run()` block, and diagnostic output is retained instead of discarded) but this project has not independently re-run the retest's predecessor attempts under the corrected harness to measure how many of them would have flipped, so no overall reliability gain is claimed or measured here, only the specific behavioural changes listed. The honest, defensible headline across both datasets combined, including round 5's genuinely multi-payload-probed result: zero patches in this project's evaluation have both passed the two-check acceptance criterion *and* been confirmed correct by trace-level inspection. Multi-payload re-probing happened in one case (round 5) but was uninformative there because the same crash most plausibly recurs regardless of payload content (Section 4.1.2); it never happened at all for the retest's three nominally accepted patches.

The corrected validator introduced in Section 4.1.2 directly addresses the specific failure mode this subsection describes, but only partially, and its own results should be read with the same care applied to every other number in this project. Replaying the round-5 trace through the corrected criterion reclassifies the false acceptance as `inconclusive_crash` rather than a pass, without needing a second manual trace inspection to catch it: the corrected criterion, not a human re-reading a log, now distinguishes a deliberate rejection from a crash on the exploit side. It does not, however, close the equivalent gap on the benign side: Section 4.1.2 documents, from a direct audit prompted by external review, that `patch_function_preserved` requires only the absence of a crash signature on the benign probe, not a correct HTTP response or a class-specific correctness check, and that 14 of 27 `confirmed_fix` verdicts in the 36-run hosted comparison returned a non-200 status on that probe without being flagged. This is a genuine improvement to the validation harness's exploit-side discrimination, not a claim that patch *quality* or benign-side correctness has been independently confirmed: applying the corrected validator to a full sixth catalogue round found zero `confirmed_fix` results (three `not_blocked`, one `inconclusive_crash`, out of four patch attempts against four confirmed exploits), and a later 36-run hosted-model comparison (Section 4.1.6) found `confirmed_fix` rates of only 4–5 of 6 per model under this same heuristic screening criterion, even with substantially more capable models than the locally hosted comparison in this section. The corrected validator makes this project's exploit-side patch-acceptance numbers more trustworthy; it does not, by itself, make patch generation succeed more often, and its benign-side check remains a screen for gross regressions, not a correctness proof.

A separate, more procedural finding reinforces the same caution: three of the four harness defects found during the patch-validation investigation (the markdown fence, the missing `app.run()` block, and the discarded diagnostic) were pure implementation bugs, not model failures. It is plausible, given their nature, that they suppressed some patch-

attempt and two-check acceptance counts in earlier rounds and comparisons, but this project did not re-run those specific earlier attempts under the corrected harness, so this remains a plausible effect of the identified bugs rather than a measured, counterfactually confirmed historical count. This is consistent with the pattern already observed for exploit confirmation in Section 4.2.1: a nontrivial share of this project's apparent capability gaps has turned out on inspection to be harness defects rather than fundamental model limitations. The corollary, given the false-acceptance finding above, is that this cuts in *both* directions (a harness defect can suppress a real success (as with the round-3 markdown-fence crashes) or manufacture a false one (as with round 5's `NameError`)) so neither a low nor a high raw count can be taken as a reliable lower bound without independent trace verification.

4.2.3 Runtime and Model-Size Findings

It is clear from Table 4.5 that confirmation rate did not track parameter count: the 7B qwen2.5-coder model outperformed both the 9B (gemma2, 0/6) and 13B (codellama, 2/6) models evaluated. This is consistent with the general finding in the wider LLM literature that code-specialised training and fine-tuning can matter more than raw parameter count for narrow, structured generation tasks, though this project's comparison (a single deterministic shot per CVE per model, with qwen2.5-coder:7b carrying prior prompt-tuning exposure the other models did not have) is a real confound and should not be read as a general-purpose model ranking. The codellama:13b and gemma2:9b runtime anomalies (respectively far slower than a larger model, and one 15–25x single-run outlier) were not root-caused in this project; they are reported as observed rather than explained by an unverified guess, consistent with this project's evidence-gating discipline applied to its own reported findings, not only to its exploit-confirmation claims.

4.2.4 Classification Reliability and the Structured-Data Advantage

It is clear from Table 4.7 that preferring an advisory's own structured CWE field over free-text keyword matching removed every classification error observed in this catalogue, which supports treating this precedence rule as a deliberate design choice rather than an arbitrary implementation detail. The one full miss under prose-only matching (CVE-2026-42208) suggests that keyword-based classification is fragile specifically when an advisory describes a vulnerability's mechanism without using the vulnerability class's own common vocabulary: a gap a structured field sidesteps entirely, since it does not depend on any particular choice of words in the prose description. The single exact-CWE mismatch, however, points to a narrower and more specific limitation than "keyword

matching is unreliable" in general: a fixed family-to-representative-CWE mapping cannot reproduce an analyst's more specific subtype judgment (here, CWE-80 assigned instead of the mapping's default CWE-79 for any detected cross-site-scripting pattern), which is a property of how the mapping was hand-encoded, not of prose matching as a technique. This distinction matters for how the result should be read: it does not establish that prose-based classification is unreliable in general, only that a coarse family mapping loses subtype precision that a structured field preserves.

4.2.5 Relationship to the AIxCC/OSS-CRS Benchmark Context

The 86% discovery / 68% patch figures introduced in Chapter 1 are DARPA's own aggregate across all seven AIxCC finalist systems combined, not a per-system figure and not specifically ATLANTIS's own result; the source material available for this review does not report an ATLANTIS-specific patch rate. Direct numerical comparison between that aggregate figure and this project's own confirmation and patch-acceptance rates would also be comparing results obtained under materially different tasks, software, and evaluation conditions: AIxCC's finalists discovered previously *unknown* bugs seeded into large, real competition codebases under a shared, adjudicated benchmark, while this project confirms and reproduces *already-published* bugs against six small, generated stand-in targets. The two results are not on the same scale or the same denominator, and no numerical comparison between them is drawn here.

What is fair to say is qualitative rather than quantitative: both AIxCC-class systems and this project treat an unverified claim as insufficient on its own and require executable evidence before a discovery or a patch is trusted. FuzzingBrain, a real AIxCC finalist system, uses a comparable staged reachability-then-triage architecture and reports that its call-graph analysis narrows a codebase from thousands of functions to a more tractable reachable subset [10], which is qualitatively consistent with this project's own reachability-filtering result; however, FuzzingBrain's own published paper does not report a specific percentage reduction figure comparable to this project's 96.1%, so no quantitative comparison between the two is drawn here, only the shared qualitative design choice.

4.2.6 Limitations, Trade-offs, and Sources of Uncertainty

The exploit-artifact generation results above were obtained against a generated Flask stand-in reproducing each advisory's described bug pattern, not the actual named upstream package; a confirmed exploit therefore demonstrates that the described pattern is exploitable as reproduced, not that the real package is exploitable when actually installed and run. This limitation is now partially, not fully, closed: Section 4.1.3 demonstrates the same evidence-gating discipline against a real, pinned-version PyPI package (CVE-2026-23949) rather than a reproduction, reaching this project's strongest evidence tier

among the Python-based main catalogue's cases, but this remains one case chosen for tractability out of six, not a claim that the remaining five could be validated the same way without comparable additional effort. The reachability result avoids the reproduction-vs-real-package limitation by a different route, since it operates on the real, cloned upstream free5GC source, and its original compile-only bound (`go build` succeeding proves syntactic validity and compile-compatibility, not runtime correctness) has since been closed for all four handlers rather than left standing: a harness satisfying only UDR's own MongoDB and NRF dependencies (Section 3.8.2), first minimally and then via a fuller Docker deployment with a real NRF, confirmed, over real HTTP against both the vulnerable and fixed commits, that the fix blocks all four handlers' exploitable behaviour (an unauthorized single-record read, an unauthorized collection-wide read, an unauthorized write, and an unauthorized delete) while preserving legitimate function: one of three real-upstream, runtime-confirmed cases alongside `jaraco.context` and OpenEMR (Section 4.1.4), and the only one where all of a component's reachable handlers, not a single case, reached this standard. Deploying and testing the full free5GC core network stack (AMF, SMF, and the rest) remains out of scope, as does OAuth2 enforcement, which the fuller deployment's real NRF enables by default and which this harness explicitly disabled to reach the handlers under test. The multi-model comparison's single-shot, non-repeated design (one run per CVE per model, even with sampling pinned) means individual model rankings in Table 4.5 should be read as one data point rather than a statistically powered result: this was an explicit, acknowledged design trade-off, made to keep the eight-model, 48-run comparison tractable within the project's timeframe rather than to claim more statistical confidence than the design supports.

4.3 Summary and Implications

Overall, these findings suggest that an evidence-gated, dynamically-confirmed pipeline of this kind can produce useful confirmation evidence on a small, fixed catalogue of already-disclosed vulnerabilities, but that patch generation and validation remain substantially harder to achieve reliably with locally hosted, moderately sized open-weight models than exploit confirmation does, and harder still to trust once a validation criterion is checked by trace-level inspection rather than taken at its own flag. No numerical comparison to AIxCC finalist systems is drawn here, per the discussion in Section 4.2.5; the two bodies of work share only a qualitative commitment to executable evidence over unverified claims. This may have implications for how future work in this space allocates effort: rather than treating exploit confirmation and patch generation as symmetric problems solvable by the same prompt-and-model combination, the results here suggest patch generation may benefit from separate, class-specific tuning, or from a larger/more capable model tier, before it can be trusted as a validated pipeline output rather than an experimental one.

For practitioners considering a similar system, the reachability-filtering result offers a more immediately transferable takeaway: narrowing an LLM's analysis scope via real, AST-based call-graph reachability (rather than asking a model to reason over an entire source file) produced a large, real reduction in analysis scope (96.1%) and an initial patch independently verified to compile against the genuine upstream dependency graph for all four reachable handlers, without at that stage requiring the more resource-intensive dynamic confirmation that the main pipeline relies on. This shows reachability filtering and dynamic confirmation are not mutually exclusive: a minimal, dependency-scoped harness (Section 3.8.2) first added runtime evidence for one filtered-in handler (DELETE) without deploying the full upstream stack, and a fuller Docker-based deployment subsequently extended the same dynamic standard to all four reachable handlers, without ever needing the rest of the free5GC core network stack (AMF, SMF, and the rest).

Chapter 5

Conclusion

5.1 Summary of the Project

This project set out to extend an existing exploit-confirmation prototype so that suspected vulnerabilities could be confirmed and candidate fixes evaluated with genuine evidence, rather than accepted on the strength of a plausible-sounding generated artifact alone. Two complementary capabilities were built and evaluated toward this goal. The main pipeline advanced a CVE from advisory intake through classification, exploit-artifact generation, dynamic execution, self-improvement, and patch generation and validation, evaluated across five rounds against a six-CVE catalogue and, separately, across eight locally hosted models. A second, independent capability ported and extended an earlier reachability-filtering prototype to operate on real, disclosed free5GC source, narrowing a real 102-function file to 4 relevant functions and verifying a generated patch against the genuine upstream project; all four of those handlers (CVE-2026-40248) were subsequently runtime-confirmed, first against a minimal MongoDB-only stand-in and then against a fuller deployment with a real free5GC NRF, rather than compiled but never executed, making free5GC the project's most complete demonstration of the full exploit-confirmation-and-patch-validation methodology against a real, disclosed, compiled upstream vulnerability.

5.2 Objective Closure

Table 5.1 maps this project's stated objectives (Chapter 1) directly onto the evidence that closes each one, so the status of each is stated once, explicitly, rather than left for a reader to reconstruct from Chapter 4's narrative.

Table 5.1: Objective-by-objective closure.

Objective	**Status**	**Evidence**
Dynamic exploit confirmation (six-CVE catalogue)	Met	All six vulnerability classes confirmed at least once across five Ollama rounds; every one of six hosted Claude models confirmed all six exploits (36/36) in the later comparison (Section 4.1.6).
Patch generation and validation	Partially met	The real-upstream `jaraco.context` (Section 4.1.3) and OpenEMR (Section 4.1.4) cases and the 6 recoverable, content-verified synthetic patches (Section 4.1.2) are confirmed correct; the broader 36-run comparison's `confirmed_fix` rate (4–5/6 per model) remains heuristic patch-screening, not confirmed correctness, at that scale.
free5GC reachability filtering & patch verification	Partially met	102 real functions reduced to 4 reachable (96.1%), all classified correctly; all four generated patches verified to compile against the real, live-cloned upstream dependency graph, and all four handlers (CVE-2026-40248) subsequently runtime-confirmed via a fuller Docker deployment with a real MongoDB instance and a real free5GC NRF (Section 3.8.2): OAuth2 enforcement was disabled to reach the handlers under test, and no live HTTP exploit was run against a fully deployed free5GC core network (AMF, SMF, and the rest) (Section 3.10).
ProvTrail/SARIF integration	Partially met	SARIF ingestion plumbing (`provtrail_bridge.py`) is functional and one hand-authored JavaScript case (CVE-2024-48910) is dynamically confirmed; this is not a systematic evaluation of ProvTrail's own flagged findings against real scanned projects.
Multi-model comparison under a fixed protocol	Met	Eight locally hosted models (48 runs) and six hosted Claude models (36 runs), both under a frozen, single-protocol comparison with real cost/time accounting.

5.3 Key Findings

Eight findings stand out from the results reported in Chapter 4.

1. Dynamic exploit confirmation against local reproductions reached 5 of 6 CVEs in the best single Ollama round, and every class in the catalogue confirmed at least once across all rounds combined: evidence that the underlying generation-and-execution approach is viable across a range of vulnerability classes on the reproductions tested, not just the one it was originally tuned against, and not independent evidence about the named upstream packages themselves (Section 3.10).

2. Two of the confirmation-rate swings observed across rounds were traced to specific, fixable implementation defects rather than to model capability, underscoring that a pipeline's reported "confirmation rate" is only as trustworthy as its own surrounding plumbing, and that this distinction is worth checking explicitly rather than assumed.

3. Patch generation and acceptance lagged exploit confirmation substantially throughout the local-model rounds: at most one of six catalogue CVEs was flagged as accepted by the two-check criterion in any round; zero of forty-eight runs in the initial eight-model comparison; and three of forty-eight in a later, separate retest run after four harness bugs were fixed, none of which was tested against more than its own original payload.

4. Most important for how the first three findings should be read: the validation criterion behind that "passing" flag was itself found, by trace-level inspection, to accept a patch for the wrong reason at least once: a generated patch whose vulnerable route crashed on a missing import was flagged as validated identically to a patch that deliberately blocked the exploit, because the automated criterion cannot tell the two apart. This is reported as a genuine capability gap discovered during this evaluation, not a pre-existing flaw papered over, and it means every raw pass/fail count in this project, not only the patch-validation ones, should be read as what the harness reported, subject to the same kind of trace-level check that surfaced this specific case, rather than as an automatically trustworthy final verdict.

5. Once that criterion was corrected into a four-outcome verdict (Section 4.1.2) and applied at greater scale (a full sixth Ollama catalogue round and a 36-run comparison across six hosted Claude models (Section 4.1.6)) patch generation remained the project's principal unresolved capability gap rather than a solved one: even the strongest hosted models reached only 4–5 of 6 `confirmed_fix` outcomes under the corrected criterion's heuristic patch-screening (itself found, on direct audit, to require only crash-absence on the benign probe rather than a correct HTTP response),

and cost tracked model tier without tracking screening-pass rate, with the cheapest hosted model tested matching the best rate at 20–42% of every other model's real cost (the ratio varies by comparison model, not a single fixed fraction).

6. One real-upstream case (Section 4.1.3) demonstrated this project's full evidence-gating discipline (real exploit, real published fix, real benign-use check) against an actual, pinned-version PyPI package rather than a generated reproduction, partially closing a limitation stated since this project's earliest results, though as one case out of six, not a systematic replacement for the reproduction-based catalogue.

7. A second real-upstream case, structurally unrelated to the first, transferred the same evidence-gating method to a different application domain and language: OpenEMR's GHSA-q366-cv5v-83w8 (Section 4.1.4) was confirmed against the real, unmodified application, with the legitimate-use check verified against exact expected content rather than crash-absence alone, reaching the same content-verified evidence tier as `jaraco.context` on a target chosen specifically because it had no counterpart in the Python-packaging or telecom/5G contexts already covered.

8. The same standard was independently reached for a third, structurally different real-upstream case: the free5GC reachability work's compile-time-only limitation was closed for all four of its handlers (CVE-2026-40248), first for the DELETE handler by building and running the real, cloned `free5gc/udr` module against a real MongoDB instance and a minimal stand-in NRF, then for the remaining three handlers against a fuller Docker deployment with a real free5GC NRF. The fuller deployment found the same root cause to be more severe than the DELETE case alone suggested: the vulnerable commit permits an unauthorized read of a single record, an unauthorized read of the entire subscription collection, and an unauthorized write, in addition to the already-confirmed unauthorized delete, with the real fix commit closing all four while preserving legitimate behaviour: this project's only case combining reachability filtering, exploit confirmation, and patch validation against a real, compiled, non-Python upstream target, and the only case where every one of a component's reachable handlers reached this standard.

5.4 Contributions

This project's contributions are: (1) a dynamic-execution and self-improvement extension to an existing exploit-confirmation pipeline, evaluated with real subprocess and HTTP evidence rather than static description; (2) a two-check, evidence-gated patch-acceptance mechanism, and an honest, quantified account of where it does and does not currently succeed, including a corrected four-outcome verdict (Section 4.1.2) built specifically in

response to a confirmed false acceptance the original criterion could not detect on its own, and itself subsequently audited to state precisely what its own benign-route check does and does not establish; (3) a reachability-filtering capability applied for the first time to real, disclosed free5GC source, including compile-time verification against the actual upstream project's dependency graph for all four handlers and runtime confirmation, for all four (CVE-2026-40248), first against a real, locally hosted MongoDB backend and a minimal stand-in NRF, then against a fuller Docker deployment with a real free5GC NRF that found the underlying bug to be a full CRUD bypass (unauthorized read, collection-wide read, write, and delete) rather than a single-record issue: a result neither this project's own predecessor nor any prior version of this pipeline had previously demonstrated for a Go target, and this project's most complete real-system case, combining reachability filtering, exploit confirmation, and patch validation against a single disclosed vulnerability in genuine upstream source; (4) a reproducible, sampling-pinned, eight-model comparison quantifying how confirmation rate and patch-acceptance rate vary with model choice under an identical protocol, and a second, 36-run comparison extending the same protocol to six hosted Claude model snapshots with real, metered per-call cost and token accounting; and (5) two further real-upstream validation cases demonstrating this project's full exploit-confirmation-and-patch-verification methodology against actual, unmodified software rather than a generated reproduction: an actual, pinned-version, publicly distributed PyPI package (Section 4.1.3), and a real, disclosed OpenEMR advisory confirmed against an unmodified deployment in a second, unrelated application domain (Section 4.1.4).

5.5 Limitations

The results in this project should be read within stated bounds. The main pipeline's exploited targets are generated reproductions of each advisory's described bug pattern for five of the six main-catalogue CVEs; a confirmed exploit for those five demonstrates the described pattern is exploitable as reproduced, not that the real package is exploitable when installed and run. One case, CVE-2026-23949, is now validated against the real, pinned-version package instead (Section 4.1.3), but this remains a single case selected for tractability, not evidence that the remaining five reproductions could be replaced the same way without comparable additional effort. The free5GC reachability result avoids the reproduction-versus-real-package limitation by a different route, operating on real, cloned source; all four of its handlers' patches were verified to compile against the real dependency graph, and all four (CVE-2026-40248) were additionally confirmed at runtime, using harnesses satisfying only UDR's own MongoDB and NRF dependencies, without deploying the rest of the core network stack (AMF, SMF, and the rest) or enabling OAuth2 enforcement on the real NRF used in the fuller deployment: giving this project's strongest patch-validation evidence among its compiled, real-upstream reachability work,

and the only case where every reachable handler in a component reached this standard. The eight-model local comparison used a single deterministic shot per CVE per model rather than repeated trials, and the strongest-confirming model in that comparison carried prior prompt-tuning exposure the others did not, a real confound acknowledged rather than hidden; the later 36-run hosted-model comparison (Section 4.1.6) shares the same single-shot design and adds its own: the Claude CLI backend exposes no temperature or seed control at all, so even "deterministic in intent" is a weaker guarantee there than the Ollama comparison's pinned sampling. Patch generation, the project's clearest capability gap, was evaluated with every model this project had access to across both comparisons, including several hosted models substantially larger than the local eight-model set, and the gap persisted at 4–5 of 6 `confirmed_fix` outcomes even at that larger scale; whether per-class prompt tuning specifically, rather than raw model scale, would close it further was not tested.

5.5.1 Final evidence freeze and interpretation

After the comparisons reported above, three further model-access methods were evaluated against the same free5GC, OpenEMR, and six-CVE-catalogue targets described throughout this chapter: a direct Claude API gateway (AIxTech), the OpenAI API (23 model identities), and the Codex CLI (18 model identities): together a substantially larger volume of model–target pairs than the original Ollama and Claude CLI comparisons, but evaluated after this chapter's narrative was largely written and so summarised here rather than integrated chapter-by-chapter; the full per-model results are reported in the project repository (`docs/RESULTS_BY_ACCESS_METHOD.md`, `docs/FINAL_EVIDENCE_SUMMARY.md`), not reproduced in full here. Zero genuine bypasses were confirmed by any of these later methods against any already-validated patch, consistent with every earlier comparison in this chapter. The final evidence freeze that closed this later evaluation makes two interpretation rules explicit. First, the free5GC headline result is a full pipeline validation outcome: in the harness, four reachable handlers are tested, but one handler change is model-generated and three are deterministic reachability materialisations. Thus the OpenAI sweep's 18/23 figure is not an 18/23 rate of independently model-generated four-handler patches. Second, the catalogue XSS bypass result is based on browser execution rather than the legacy response-marker oracle. Thirty-three historical raw-positive proposals were replayed in headless Chrome with dialog and error instrumentation; zero produced a genuine browser execution signal. The raw marker result remains preserved for audit.

A further, separate follow-up closed one of the limitations stated throughout this chapter and in Chapter 5: every free5GC runtime result reported above runs with OAuth2 enforcement explicitly disabled on the NRF, leaving open whether a real, OAuth2-enforced deployment changes CVE-2026-40248's severity. An opt-in compose override enabled

OAuth2 on the NRF without altering the default stack any other result in this project relies on, and re-ran three of the original five runtime checks (collection-wide read, single-record read, and the benign path; the unauthorized-write and confirmed-delete checks were not ported, for time reasons) against both the vulnerable and the real-fix-patched UDR image, under two conditions: no credentials at all, and a valid, correctly-scoped bearer token minted with the NRF's own private key for a real, currently-registered network-function identity (not a full client-credentials HTTP round-trip, which was out of scope for this one check). With no credentials, both builds correctly returned HTTP 401 before the underlying CWE-285 bug could matter at all. With a valid token, the vulnerable build's collection-wide and single-record leaks reproduced exactly as in every OAuth2-disabled result, and the real fix commit closed both while preserving the legitimate path, also exactly as before. OAuth2 enforcement and the code-level fix are therefore independent, complementary layers, not substitutes for each other: OAuth2 narrows the threat model from "any anonymous HTTP client" to "any entity holding a validly-scoped token," which matters substantially against a pure outsider, but provides no additional protection against this specific authorization-bypass bug once that bar is cleared, so the real fix remains necessary even in a deployment that enforces OAuth2. Full method, the two real implementation complications found and resolved before trusting this result (a second, OAuth2-only authorization layer on the write path requiring a real registered NF identity rather than a fabricated one; and a concatenated-looking response body independently confirmed genuine via its raw `Content-Length` header), and the raw per-request evidence are in `docs/FREE5GC_OAUTH2_ENFORCEMENT_RESULTS.md` and `reports/reachability/free5gc_oauth2_enforcement.json`.

The targeted repeatability follow-up used unique prompt markers for Claude and the AIxTech gateway. All 12 AIxTech calls completed with token and duration measurements; nine of twelve Claude CLI calls completed, while three Opus calls were retained as access failures. A subsequent controlled OpenAI repeatability attempt recorded the requested temperature and seed for all twelve runs, but the locally supplied API key was rejected with HTTP 401 before generation. Those attempts are therefore environment-limited evidence, not model-quality failures. The exact reports, commands, and final tag are recorded in the repository's reproduction record and evidence ledger.

5.6 Future Work

Four directions follow directly from the limitations above.

1. Patch generation would benefit from the same bounded, per-class tuning process already applied successfully to exploit-artifact generation in this project, repeated one vulnerability class at a time rather than attempted generically across all six at

once; every model in the 36-run comparison was recorded as `not_blocked` on the same one class (Cross-Site Scripting) under the original two-check criterion, which motivates starting there, though this should be read as a legacy screening result rather than proof every model wrote an incorrect XSS patch: the content-verified reassessment (Section 4.1.2) found at least one of these recorded XSS failures (`claude-opus-4-8`'s surviving patch) was actually a false negative in the bypass-check's own escaping logic, not a genuine patch defect.

2. The free5GC reachability work's runtime validation (Section 3.8.2), now confirmed for all four of its handlers (CVE-2026-40248) under a fuller deployment with a real NRF, has since been extended with OAuth2 enforcement turned on (Section 5.5.1): the headline question of whether OAuth2 changes this CVE's exploitability by a credentialed caller is answered (it does not), but that follow-up only ported 3 of the original 5 runtime checks and minted its bearer token directly with the NRF's own private key rather than through a live client-credentials HTTP round-trip against NRF's own `/oauth2/token` endpoint. Porting the remaining unauthorized-write and confirmed-delete checks to the OAuth2-enabled harness, implementing the full client-credentials flow rather than a directly-minted token, and standing up additional network functions (AMF, SMF, and the rest) to test behaviour beyond UDR's own component boundary all remain future work, unifying the reachability engine's real-source analysis with the main pipeline's dynamic-confirmation discipline at greater scope.

3. The main pipeline's exploited targets could be moved from generated reproductions toward the actual named upstream packages for a larger subset of the catalogue than the one case demonstrated here, extending Section 4.1.3's methodology to the remaining five classes.

4. The hosted-model comparison could be extended with sampling controls once the Claude CLI (or an equivalent hosted interface) exposes them, and repeated across multiple shots per CVE per model to move from one data point per model toward a statistically powered ranking.

Bibliography

[1] "The Hugging Face incident and the road ahead," OpenAI, Aug. 26, 2026. https://openai.com/index/hugging-face-incident-and-the-road-ahead/ (accessed Sep. 22, 2026).

[2] E. Schluntz and B. Zhang, "Building effective AI agents," Anthropic Engineering, Dec. 19, 2024. https://www.anthropic.com/engineering/building-effective-agents (accessed Sep. 22, 2026).

[3] E. Weiss and N. Khayet, "Can AI weaponize new CVEs in under 15 minutes?," Efi Weiss, Aug. 21, 2025. https://valmarelox.substack.com/p/can-ai-weaponize-new-cves-in-under (accessed Sep. 22, 2026).

[4] Defense Advanced Research Projects Agency, "AIxCC Final Competition Results," DARPA, 2025. https://www.darpa.mil/news/2025/aixcc-results (accessed Sep. 25, 2026).

[5] T. Kim et al., "ATLANTIS: AI-driven threat localization, Analysis, and Triage intelligence system," arXiv (Cornell University), Sep. 2025, doi: 10.48550/arxiv.2509.14589.

[6] "OSS-CRS — Cyber Reasoning Systems for Open source software | OSS-CRS." https://oss-crs.openssf.org/ (accessed Sep. 22, 2026).

[7] The Linux Foundation, "World's Leading Open Source Mobile Packet Core, free5GC, Moves Under Linux Foundation to Provide Open Source Alternatives Across 5G Deployments," THE LINUX FOUNDATION, Sep. 16, 2024. https://www.linuxfoundation.org/press/worlds-leading-open-source-mobile-packet-core-free5gc-moves-under-linux-foundation-to-provide-open-source-alternatives-across-5g-deployments (accessed Sep. 22, 2026).

[8] M. C. Fanning and L. J. Golding, "Static Analysis Results Interchange Format (SARIF) version 2.1.0." Mar. 27, 2020. [Online]. Available: https://docs.oasis-open.org/sarif/sarif/v2.1.0/os/sarif-v2.1.0-os.html

[9] N. Korda and G. Evron, "OpenAnt: LLM-Powered Vulnerability Discovery Through Code Decomposition, Adversarial Verification, and Dynamic Testing," arXiv preprint arXiv:2606.19149 (2026). https://arxiv.org/abs/2606.19149

[10] Z. Sheng, Q. Xu, J. Huang, M. Woodcock, H. Huang, A. F. Donaldson, G. Gu, and J. Huang, "All You Need Is A Fuzzing Brain: An LLM-Powered System for Automated Vulnerability Detection and Patching," arXiv preprint arXiv:2509.07225, 2025. https://arxiv.org/abs/2509.07225. Competition repository (organisation `fuzzingbrain`, formerly listed as `o2lab`): https://github.com/fuzzingbrain/afc-crs-all-you-need-is-a-fuzzing-brain.

[11] Ollama. https://ollama.com/ (accessed Sep. 2026).

[12] W. Wang, W. Ma, Q. Hu, Y. Zhang, J. Sun, B. Wu, Y. Liu, G. Xu, and L. Jiang, "VulnRepairEval: An Exploit-Based Evaluation Framework for Assessing Large Language Model Vulnerability Repair Capabilities," arXiv preprint arXiv:2509.03331, Sep. 2025. https://arxiv.org/abs/2509.03331.

[13] Z. Yu, W. Shi, X. Sun, Z. Feng, M. Xu, and X. Xing, "Patch Validation in Automated Vulnerability Repair," arXiv preprint arXiv:2603.06858, Mar. 2026. https://arxiv.org/abs/2603.06858.

[14] D. Gezgin, A. Das, S. Kim, Z. Huang, N. Stojkovic, and C. Wang, "PoC-Gym: Towards More Reliable LLM-Assisted Proof-of-Concept Exploit Generation," arXiv preprint arXiv:2602.04165, Feb. 2026. https://arxiv.org/abs/2602.04165.

[15] R. Fang, R. Bindu, A. Gupta, and D. Kang, "LLM Agents can Autonomously Exploit One-day Vulnerabilities," arXiv preprint arXiv:2404.08144, Apr. 2024. https://arxiv.org/abs/2404.08144.

[16] Z. Qi, F. Long, S. Achour, and M. Rinard, "An Analysis of Patch Plausibility and Correctness for Generate-and-Validate Patch Generation Systems," in Proceedings of the 2015 International Symposium on Software Testing and Analysis (ISSTA), 2015. doi: 10.1145/2771783.2771791.

[17] C. Zhang et al., "SoK: DARPA's AI Cyber Challenge (AIxCC): Competition Design, Architectures, and Lessons Learned," arXiv preprint arXiv:2602.07666, Feb. 2026. https://arxiv.org/abs/2602.07666.

[18] Y. Zhu, A. Kellermann, A. Gupta, P. Li, R. Fang, R. Bindu, and D. Kang, "Teams of LLM Agents can Exploit Zero-Day Vulnerabilities," arXiv preprint arXiv:2406.01637, Jun. 2024. https://arxiv.org/abs/2406.01637.

[19] S. Ullah, P. Balasubramanian, W. Guo, A. Burnett, H. Pearce, C. Kruegel, G. Vigna, and G. Stringhini, "From CVE Entries to Verifiable Exploits: An Automated Multi-Agent Framework for Reproducing CVEs," arXiv preprint arXiv:2509.01835, Sep. 2025. https://arxiv.org/abs/2509.01835.

[20] Q. Liu, W. Zhang, M. Jiang, L. Wu, and Y. Zhou, "Characteristics, Root Causes, and Detection of Incomplete Security Bug Fixes in the Linux Kernel," arXiv preprint arXiv:2511.17799, Nov. 2025. https://arxiv.org/abs/2511.17799.

Appendix A

Reproducibility Details

A.1 Full CVE Catalogue

Table A.1: All nine CVEs used across this project's evaluation: six in the main catalogue (Chapter 4), two in the free5GC work (Chapter 3), and one in the ProvTrail integration (Chapter 3).

CVE	**Package**	**Class**	**CWE**	**Real source or reproduction**
CVE-2026-42208	litellm	SQL Injection	CWE-89	Reproduction (`target_app.py`)
CVE-2026-27602	modoboa	OS Command Injection	CWE-78	Reproduction
CVE-2026-23949	jaraco.context	Path Traversal	CWE-22	Reproduction
CVE-2026-78683	nltk	Insecure Deserialization	CWE-502	Reproduction
CVE-2026-54729	dssrf (npm)	SSRF	CWE-918	Reproduction
CVE-2026-46492	md-fileserver	XSS	CWE-80	Reproduction
CVE-2026-40246	free5GC/udr	Improper Auth.	CWE-285	Reproduction (`free5gc_lab/`)
CVE-2026-40248	free5GC/udr	Improper Auth.	CWE-285	Real upstream source; 4 handlers compile-verified, 4 runtime-confirmed (fuller deployment)
CVE-2024-48910	dompurify (npm)	Prototype Pollution	CWE-1321	Reproduction (`provtrail_js_lab/`)

Every entry above was independently verified against GitHub's advisory API or commit API before use, not taken from a search-result summary. The table contains four distinct groups, not one uniform set: six CVEs (litellm through XSS) evaluated as local Python reproductions in the main catalogue (Chapter 4); CVE-2026-40246 evaluated as a separate, hand-written Go reproduction (`free5gc_lab/`) with no fix commit documented for it in this project's own records; CVE-2024-48910 demonstrated as a hand-authored JavaScript reproduction; and CVE-2026-40248, the one case where the real, cloned upstream `free5gc/udr` source itself was analysed and patched, with its own documented fix commit: compile-verified for all four reachable handlers, and additionally runtime-confirmed for all four under a fuller Docker deployment (Section 3.8.2, Appendix A). CVE-2026-40246 and CVE-2026-40248 are related (the same bug class in the same source file family) but were evaluated with different artifacts under different evidence standards, and should not be read as one shared source-file-and-fix-commit experiment. CVE-2026-23949 additionally carries a second, separate evaluation not reflected in its table row above: alongside its main-catalogue reproduction, its real, pinned-version PyPI package (`jaraco.context`, versions `5.3.0` and `6.1.0`) was independently validated against a real exploit and the real published fix (Section 4.1.3); the two evaluations use different artifacts under different evidence standards and are reported separately throughout Chapter 4, exactly as CVE-2026-40246 and CVE-2026-40248 are.

A.2 free5GC Runtime Validation Evidence Bundle

The minimal, single-handler (DELETE) runtime validation of CVE-2026-40248 described first in Section 3.8.2 and its results is reproducible from the following artifacts, all committed to the `fyp` remote (`https://github.com/J-Dheeraj/Final-Year-Project`) under `free5gc_runtime_case/`, at commit `70281e5` (which supersedes commit `f09de67`, itself a correction of an initial CVE-identifier mislabelling introduced at commit `4524193`; both corrections are documented in `docs/FREE5GC_RUNTIME_VALIDATION_PLAN.md`, and the harness was re-run after each to confirm an identical result before this report was written):

Table A.2: free5GC runtime validation evidence bundle (`free5gc_runtime_case/`).

File	Contents
`run_harness.py`	The harness itself: builds the vulnerable and fixed commits, starts the stub NRF, drives the seed/malicious-delete/benign-delete request sequence, and writes every file below.
`stub_nrf/main.go`	The minimal HTTP/2-cleartext (h2c) stand-in NRF, built with Go's standard h2c library, required only because free5GC's own NRF client negotiates h2c unconditionally and rejects an HTTP/1.1 responder.
`udrcfg.yaml`	The real UDR configuration used for both builds (local MongoDB URL, local NRF URI, cleartext SBI scheme).
`manifest.json`	Structured record of both runs: commit hashes, component/handler identifiers, services used, and the full per-run result objects.
`verdict.json`	The pipeline's own final verdict: whether the exploit was blocked, whether legitimate behaviour was preserved, the overall four-outcome verdict, and whether the vulnerable run confirmed the exploit.
`patch.diff`	The real, unmodified diff between the vulnerable and fixed commits for the affected source file.
`vulnerable_response.log`, `patched_response.log`	Full per-run HTTP status/body evidence for every request in the sequence (seed, pre-check, malicious delete, post-malicious check, reseed, benign delete, post-benign check).
`vulnerable_process.log`, `patched_process.log`	Raw stdout/stderr from each UDR process, including the diagnostic NRF-registration retry evidence that led to the stub-NRF fix.
`build_vulnerable.log`, `build_patched.log`	Full build output for both commits.
`vulnerable_commit.txt`	The exact vulnerable commit hash used (the fix commit's direct parent).

`docs/FREE5GC_RUNTIME_VALIDATION_PLAN.md` in the same repository documents the case selection rationale, the exact vulnerable/fixed commit hashes, two genuine implementation blockers found only by running the harness for real rather than by reading source (UDR's NRF registration being a blocking infinite retry rather than "logged and non-fatal" as an earlier, shallower reading concluded; and the first, HTTP/1.1 stub NRF attempt

failing with a client-side `http2: frame too large` error before the h2c requirement was identified), and the CVE-identifier correction referenced above. `docs/PROJECT_STATUS.md`'s dated session-log entries for 2026-09-27 record the same work chronologically, including the correction, as this project's established multi-session coordination record.

The fuller-deployment extension covering all four handlers, described in the second half of Section 3.8.2, is tracked separately on branch `free5gc-full-deployment` (commit `e59ee4e`), kept isolated from the single-handler bundle above so the two remain distinguishable. Its evidence bundle, under `free5gc_full_deployment/evidence/`, follows the same schema: `manifest.json` and `verdict.json` record all four handlers' vulnerable- and patched-build outcomes (`overall_verdict: "confirmed_fix_all_four_handlers"`), `vulnerable_response.json` and `patched_response.json` record the full raw HTTP status, headers, and body for every request (including the collection-GET and single-PUT requests whose raw response bytes are the direct evidence for the stronger CRUD-bypass finding), and `vulnerable_process.log`/`patched_process.log` record the real UDR container logs, including successful registration with the real free5GC NRF. `docs/FREE5GC_FULL_DEPLOYMENT_RESULTS.md` on that branch documents the deployment (real, official MongoDB and NRF images; a custom-built UDR image compiled from the same pinned commits as the single-handler case; OAuth2 explicitly disabled on the NRF, stated as an explicit limitation rather than omitted) and a caught-and-corrected mistake (an unrelated case study's commit hashes were briefly, and unsuccessfully, used for this build before the correct ones were substituted). This branch has not been merged to `main`; this report version, which does incorporate the fuller-deployment finding throughout, is itself kept on that branch pending a separate decision on whether it should replace the single-handler version. For precision: commit `e59ee4e` is where the fuller-deployment evidence bundle and its harness were established; this report source and its compiled PDF (`free5gc_full_deployment/FYP_Report_main_upgraded.tex`, `.pdf`) were committed afterward, at commit `16d9b63`, on the same branch.

A.3 Model and Sampling Configuration

Table A.3: Exact configuration values used across the live-LLM evaluation (Chapter 4), read directly from `cve_pipeline.py`.

Parameter	Value
Default local model	`qwen2.5-coder:7b` (env override: `OLLAMA_MODEL`)
Sampling (round 5 onward)	`temperature=0`, `seed=42`, pinned specifically because round 1 vs. round 2 of the live-LLM run were not directly comparable without it (identical CVE and prompt flipped confirmed/unconfirmed between runs)
Per-call generation timeout	180 seconds
Self-improvement iteration bound	3
Ollama host	`http://127.0.0.1:11434` explicitly (fixed after an IPv4/IPv6 `localhost` ambiguity was found to silently flip availability checks)
Eight-model comparison set	`qwen2.5-coder:1.5b/3b/7b`, `llama3.1:8b`, `mistral:7b`, `gemma2:9b`, `codellama:13b`, `deepseek-coder-v2:16b`
Hosted Claude comparison set	`claude-haiku-4-5`, `claude-sonnet-4-6`, `claude-sonnet-5`, `claude-opus-4-6`, `claude-opus-4-7`, `claude-opus-4-8`; no temperature/seed control exposed by the CLI
Hosted comparison generation timeout	300 seconds (raised from the 180-second default specifically for two of the six models, found to genuinely exceed the shorter budget on realistic prompts)

Exact prompt wording is versioned directly in `cve_pipeline.py` and `src/pipeline/patch.py` in the source repository (Appendix D); it is not reproduced verbatim here for space. The pinned commit's source files document the prompts as they stand in that final, pinned snapshot of the pipeline; the prompts changed across the five development rounds described

in Chapter 4, and the exact wording used in any one specific earlier round is not separately preserved outside that round's own commit history. Table A.4 maps each round to the commit(s) where its own protocol and any code changes it depended on were introduced, so an examiner can check out the closest recorded code and prompt state for a specific round rather than only the final pinned snapshot; as that table notes, round 5's exact execution working tree was not independently captured, so "closest recorded" is not a guarantee of an exact match for that round specifically.

Table A.4: Round-to-commit mapping for the live-LLM catalogue runs (Chapter 4).

Round	Commit(s)	What changed at this point
1	`390c316`	First full six-CVE catalogue run under the frozen prompt; `qwen2.5-coder:7b` via local Ollama.
2	`4c1a386`	Fixed a schemeless-URL bug found in round 1's PoCs; re-ran the full catalogue.
3	`5ecd4d0` `6f932b4`	Ran with Stage 3.7 (self-improvement) newly wired to the live backend; `6f932b4` documents the markdown-fence and stale-exit-code bugs this round surfaced, before they were fixed.
4	`b05b45d` `9711fe5`	`b05b45d` fixes the two bugs found in round 3; `9711fe5` is the round-4 catalogue run itself, with the first patch accepted by the two-check criterion.
5	`348eea0` (code) `f5958d0` (results)	Ran after `56449a2` merged a parallel session's independent live-LLM work (multi-payload validation, pinned sampling) into `main`. `348eea0` is the parent of the results commit and the closest recorded pre-results code snapshot; the preserved raw log does not itself record a commit hash or a clean-working-tree check at execution time, so this is the closest available snapshot, not an independently verified exact match. `f5958d0` ("round 5 - full catalog on reconciled main, 5/6 confirmed") is the commit that adds all six JSON reports, `RAW_LOG.txt`, and this round's section of the live-run record.
6	`78fbaf0` (validator) `fd3d22f` (relabel)	`78fbaf0` introduces the corrected validator (Section 4.1.2) and the real-upstream CVE-2026-23949 case (Section 4.1.3). This round was launched intending a Claude Code CLI comparison, silently ran on Ollama instead due to an expired CLI session, and was relabelled as a sixth Ollama round in `fd3d22f` once the mislabelling was found by auditing `generation_backend` against the run's own stated protocol.
:	`6fa5a3a` `6f159d6` `139240d` `289d8f3` `2075671`	The 36-run hosted-Claude comparison (Section 4.1.6) is not a numbered round of the main six-CVE catalogue sequence; these five commits add real cost/token capture, the silent-fallback fix, the extended-timeout option, the agentic-tool/session-persistence fix, and the completed 36-cell results, in that order.

An examiner wanting a specific round's prompt should check out that round's commit from Table A.4 rather than relying on the pinned snapshot's current source, keeping in mind the round-5 caveat stated there.

A.4 Hosted Claude Comparison: Per-CVE Cost and Time

Table A.5 gives the individual, per-CVE cost and wall-clock time underlying Table 4.6's per-model totals in Section 4.1.6. Cost is the sum of that run's generation, self-improvement, and patch-generation calls, read directly from the Claude CLI's own `total_cost_usd` field per call, not estimated. Time is the pipeline's own recorded end-to-end wall-clock duration for that CVE, including advisory/analysis/probe stages, dynamic execution, and patch validation, not model generation time alone.

Table A.5: Per-CVE cost (USD) and wall-clock time (s) for every model–CVE cell in the hosted Claude comparison (36 cells total).

Model	**CVE**	**Verdict**	**Cost**	**Time (s)**
Haiku 4.5	42208	confirmed_fix	0.26	112.3
Haiku 4.5	27602	confirmed_fix	0.26	119.0
Haiku 4.5	23949	confirmed_fix	0.24	90.6
Haiku 4.5	78683	confirmed_fix	0.55	323.0
Haiku 4.5	54729	confirmed_fix	0.26	120.4
Haiku 4.5	46492	not_blocked	0.24	84.3
Sonnet 4.6	42208	confirmed_fix	0.73	130.3
Sonnet 4.6	27602	confirmed_fix	0.34	80.3
Sonnet 4.6	23949	confirmed_fix	0.74	130.8
Sonnet 4.6	78683	not_blocked	0.80	173.5
Sonnet 4.6	54729	confirmed_fix	1.14	257.8
Sonnet 4.6	46492	not_blocked	0.74	141.0
Sonnet 5	42208	confirmed_fix	0.68	104.0
Sonnet 5	27602	confirmed_fix	0.69	99.8
Sonnet 5	23949	confirmed_fix	0.65	88.6
Sonnet 5	78683	not_blocked	0.87	123.8
Sonnet 5	54729	confirmed_fix	0.70	134.0

Model	CVE	Verdict	Cost	Time (s)
Sonnet 5	46492	not_blocked	0.67	104.5
Opus 4.6	42208	inconclusive_crash	1.24	187.7
Opus 4.6	27602	confirmed_fix	1.47	185.2
Opus 4.6	23949	confirmed_fix	1.36	259.8
Opus 4.6	78683	confirmed_fix	1.39	285.4
Opus 4.6	54729	confirmed_fix	1.28	189.4
Opus 4.6	46492	not_blocked	1.31	237.1
Opus 4.7	42208	confirmed_fix	0.77	85.6
Opus 4.7	27602	confirmed_fix	0.70	70.5
Opus 4.7	23949	confirmed_fix	0.79	106.8
Opus 4.7	78683	confirmed_fix	1.71	150.1
Opus 4.7	54729	confirmed_fix	0.73	83.7
Opus 4.7	46492	not_blocked	0.82	128.2
Opus 4.8	42208	confirmed_fix	1.74	170.2
Opus 4.8	27602	confirmed_fix	2.05	247.2
Opus 4.8	23949	confirmed_fix	0.81	189.3
Opus 4.8	78683	confirmed_fix	2.13	249.2
Opus 4.8	54729	confirmed_fix	0.95	256.8
Opus 4.8	46492	not_blocked	1.38	258.5

CVE identifiers above are abbreviated to their final five digits (e.g. `42208` = CVE-2026-42208); the class each maps to is given in Table A.1 in Section A. Raw per-cell reports: `reports/model_comparison_pilot_2026-09-25/<model>/<CVE-ID>.json`.

Summing the 36 individually-rounded cost cells shown above gives $33.19, one cent more than Table 4.6's stated $33.18 total, which is itself one cent more than that table's own per-model cells summed ($33.17). All three figures are internally consistent, not contradictory: each is the same underlying unrounded cost, accumulated and rounded in a different order (36 roundings here, 6 roundings there, versus one rounding of the true unrounded grand total, $33.179386, verified directly against the raw per-call `cost_usd` fields). This is an ordinary cascading-rounding artifact that grows with the number of addends rounded before summing, not a data discrepancy.

A.5 Validation Criteria (Quick Reference)

- **Dynamically confirmed**: a real subprocess exit code from a real HTTP exchange, or a real compiled-harness run: never a static claim.
- **Patch accepted by the two-check criterion** (the pipeline's raw `patch_validated` field): the patch applies, the original unmodified exploit now fails against it, and a separate `/health` endpoint still returns 200. As established in Chapter 4, this flag cannot distinguish a deliberate security fix from an unrelated crash bug in the patched code, and at least one flagged case in this project's own evaluation is a confirmed false acceptance on trace-level inspection. Left unchanged in every report for continuity; superseded as this project's headline patch metric by `patch_verdict` below.
- **`patch_verdict`** (the corrected, four-outcome criterion, Section 4.1.2): one of `confirmed_fix` (exploit blocked without a detected crash, and a generic benign probe to the same route also completes without a detected crash: *not* independently confirmed to behave correctly; see Section 4.1.2 for the audited scope of this check), `regression_broke_route` (blocked, but the benign probe also shows a crash signature), `inconclusive_crash` (the exploit request itself crashed the route, so blocking cannot be attributed to a deliberate check), or `not_blocked` (the exploit still succeeds). Automatically reclassifies this project's own confirmed false acceptance as `inconclusive_crash`.
- **Correct patch**: in the strict sense (diff-similarity scoring or independent second-reader adjudication against the real upstream fix's own intent) not measured by this project anywhere. In the narrower, content-verified sense (exact expected malicious-rejection and legitimate-response content, not merely crash-absence), established for three real-upstream cases (Sections 4.1.3 and 4.1.4, plus the free5GC case) where the "patch" is the real upstream maintainers' own released fix, and separately for 6 recoverable LLM-generated patched artifacts from the main catalogue (Section 4.1.2); the remaining 30 of 36 LLM-generated artifacts from that same comparison are not covered, their patched source no longer existing to re-test.
- **Compile-verified** (free5GC, all four handlers' original evidence tier): `go build` succeeds against the real, live-cloned upstream module with the patch applied: weaker evidence than dynamic confirmation, never reported interchangeably with it, and since superseded by runtime confirmation for all four handlers (see below).
- **Runtime-confirmed** (free5GC, all four handlers, CVE-2026-40248): the real, cloned module was built, run, and exercised over real HTTP for both the vulnerable

and fixed commits, first for one handler (DELETE) against a real local MongoDB instance and a minimal stand-in NRF satisfying only UDR's own startup registration call, then for all four handlers against a fuller Docker deployment with a real free5GC NRF (OAuth2 disabled): meets the same standard as **Dynamically confirmed** above, not the weaker compile-verified tier.

A.6 Relationship to AIxCC/OSS-CRS Vocabulary (Condensed)

This project's stages line up with AIxCC/OSS-CRS vocabulary as follows, condensed from this project's own internal mapping document: Stage 1 (advisory fetch) corresponds to OSS-CRS's `prepare` phase, except that it pulls an already-known advisory rather than preparing to discover one; Stage 3.5 (exploit-artifact generation) corresponds to `build-target`, except that it generates a minimal reproduction rather than compiling the real upstream project (the single biggest structural gap versus a real CRS, per Section 3.10); Stages 2, 3, 3.6, and 3.7 together correspond to the `run` phase's triage-and-verify loop. The clearest, most defensible framing is two complementary but partial capabilities: (1) the confirm-and-patch back half of a real CRS's loop, evaluated one known CVE at a time, and (2) a real, AST-based scope-reduction front half (`src/reachability/`) proven against one real, disclosed free5GC vulnerability. Neither claims the hardest capability a full CRS needs (discovering an *unknown* bug in an unmodified codebase) which this project does not attempt.

Appendix B

Selected Trace Excerpts

B.1 The False-Acceptance Case: CVE-2026-78683, Round 4 vs. Round 5

Both traces below are quoted directly from the corresponding round's raw JSON report, reformatted only for line length.

Round 4 (plausibly a genuine fix: deliberately raised exception):

```
 File "...target_app.iter1.v0.patched.py", line 20, in vulnerable
 raise ValueError("Deserialization only allowed for safe classes")
ValueError: Deserialization only allowed for safe classes
127.0.0.1 - - [23/Sep/2026 23:28:27] "POST /vulnerable HTTP/1.1" 500 -
=== poc.py (exit 1) ===
[-] FAILED
```

Round 5 (confirmed false acceptance: unrelated crash bug):

```
 File "...target_app.iter1.v0.patched.py", line 26, in vulnerable
 safe_unpickler = SafeUnpickler(io.BytesIO(data))
 ^^
NameError: name 'io' is not defined. Did you mean: 'id'? Or did you forget to import 'io'?
127.0.0.1 - - [24/Sep/2026 09:34:51] "POST /vulnerable HTTP/1.1" 500 -
=== poc.py (exit 1) ===
[-] FAILED
```

Both were recorded by the pipeline as `patch_validated: true`. Only the first is plausibly correct; the second is a crash the validation criterion could not distinguish from a fix (Section 4.1.2).

B.2 The Markdown-Fence Crash (Round 3)

Four of round 3's five unconfirmed CVEs crashed identically because a model-generated revision's `target_app.py` still contained a literal, unstripped ```` ```python ```` fence, making the file invalid Python:

```
 File "target_app.py", line 1
 ```python
 ^
SyntaxError: invalid syntax
```

Fixed in commit `b05b45d` via a dedicated `_strip_code_fence` helper; zero of round 4 or round 5's twelve reports contain this signature, checked programmatically.

B.3 free5GC Upstream Compile Verification

Output from applying all four generated patches directly to a live clone of `github.com/free5gc/udr` at the pre-fix commit and rebuilding against its real dependency graph:

```
=== Baseline: real vulnerable commit, unmodified ===
 [baseline (unpatched)] go build ./... -> OK
Applied 4/4 patches: [...]
=== After applying auto-generated patches to the REAL module ===
 [patched (real module)] go build ./... -> OK
```

B.4 free5GC LLM-Generated Patch: Multi-Model Comparison

The free5GC result reported above uses the real upstream maintainers' own fix commit, checked out directly, not a model-generated artifact. A separate, later experiment instead asked a real LLM to *write* the fix itself, for the one handler, `HandleApplicationDataInfluenceDataSubsToNotifyGet`, that the reachability engine's deterministic rule-based patcher is already documented to under-fix: that patcher inserts only one of the two `return` statements the handler's two separate early-exit validation blocks each need, since both share the identical missing-`return` bug pattern. This makes the handler a genuine test of whether a model reasons about control flow rather than pattern-matching text.

A local model (`qwen2.5-coder:7b`, via Ollama, zero marginal cost) correctly inserted `return` after both occurrences (a strict improvement over the rule-based patch) verified at the same two evidence tiers used throughout this chapter: compiled against a live clone of the real `free5gc/udr` module, then runtime-confirmed against the same real MongoDB and free5GC NRF deployment described in Section 3.8.2, yielding `overall_verdict: "confirmed_fix_all_four_handlers"`.

This was then extended to a comparison across every model locally available and across eight named hosted Claude models, keeping the target handler, the rule-based patches for the other three handlers, and the two-tier verification method fixed, varying only the model asked to patch the one handler.
```

Table B.1: free5GC LLM-generated patch: eight locally-hosted models (Section B.4).

Model	Compiles	Applied/4	Runtime verdict
`deepseek-coder-v2:16b`	No: out-of-memory	0/4	not reached
`codellama:13b`	Yes	4/4	confirmed_fix_all_four_handlers
`gemma2:9b`	Yes	4/4	confirmed_fix_all_four_handlers
`mistral:7b`	Yes	4/4	confirmed_fix_all_four_handlers
`llama3.1:8b`	Yes	4/4	confirmed_fix_all_four_handlers
`qwen2.5-coder:7b`	Yes	4/4	confirmed_fix_all_four_handlers
`qwen2.5-coder:3b`	Yes	4/4	confirmed_fix_all_four_handlers
`qwen2.5-coder:1.5b`	Yes	4/4	**partial_or_inconclusive**

Six of eight local models compiled and were runtime-confirmed. `deepseek-coder-v2:16b` never produced a patch at all: it failed to load under this machine's available memory (a 45 GB buffer allocation request), a hardware limit, not evidence about that model's patching ability. `qwen2.5-coder:1.5b` is the more informative negative result: its patch *compiled cleanly* but did not fix the bug: instead of the needed `return` statements, it appended an unrelated, syntactically valid response line after the function's existing logic. The runtime harness caught this precisely (`collection_get_leak_fixed: false`); a compile-only check, as used for the main free5GC evidence elsewhere in this chapter, would have reported this patch as equally successful as the other seven. This is a direct, concrete illustration of why this project's reachability track treats compile verification as strictly weaker evidence than runtime confirmation.

Table B.2: free5GC LLM-generated patch: eight hosted Claude models, real metered cost (Section B.4).

Model	Compiles	Cost (USD)	Runtime verdict
`claude-haiku-4-5`	Yes	$0.2138	confirmed_fix_all_four_handlers
`claude-sonnet-5-5`	Yes	$1.1334	confirmed_fix_all_four_handlers
`claude-opus-5-5`	Yes	$0.1644	confirmed_fix_all_four_handlers
`claude-sonnet-4-6`	Yes	$0.1494	confirmed_fix_all_four_handlers
`claude-sonnet-5`	**Refused**	$1.1130	not reached
`claude-opus-4-6`	Yes	$0.2477	confirmed_fix_all_four_handlers
`claude-opus-4-7`	Yes	$0.3427	confirmed_fix_all_four_handlers
`claude-opus-4-8`	Yes	$0.3460	confirmed_fix_all_four_handlers

Seven of eight hosted Claude models (accessed via the `claude` CLI's own authenticated session rather than a separate API key, mirroring the mechanism this project's main

pipeline already uses when a raw key is not configured) compiled and were runtime-confirmed, for a combined real, metered cost of $3.71. `claude-sonnet-5` is the remaining negative result, and a different kind from `qwen2.5-coder:1.5b`'s: it is confirmed as a real, distinct model (a 1,000,000-token context window and `firstParty` provider in the CLI's own usage report), but every attempt to send it this handler's patch-generation prompt was refused by that provider's own real-time content safeguard, which flags prompts mentioning CVE/CWE/vulnerability terminology as requiring separate enrolment in a verification programme before certain security-related requests are answered. The refusal itself was still billed in full. This is reported as a finding in its own right: a model from a vendor positioning its models for security-assistance use was, on this occasion, blocked by that same vendor's own safeguard from performing exactly that kind of task.

Two further robustness issues were found and fixed directly as a consequence of this comparison, both in the shared patch-generation code path used by every model and provider in this project, not specific to any one model. First, the patch-generation prompt template was originally written for a C target and retained its C-specific framing and code-fence language even when applied to free5GC's Go source; a language-aware system prompt and fence selection, keyed on the target file's extension, was added. Second, one hosted model twice ignored an explicit "no commentary" instruction in two different ways across two separate attempts (once appending an explanation after a closing code fence, once with no fence at all) in both cases leaving its underlying fix logically correct but appending trailing prose that corrupted the surrounding Go source and caused a compile failure unrelated to the fix itself. This was resolved by truncating the model's response at the target function's own final, unindented closing brace unconditionally, rather than relying on fence detection, since every target in this comparison is syntactically a single complete function.

This comparison does not control for training-data memorisation: the real, public fix commit may appear in some models' training data, and a model reproducing it from memory is, in this experimental design, indistinguishable from one genuinely re-deriving it from the function body and vulnerability description alone. Each model was also run once; `qwen2.5-coder:7b` was observed to produce a textually different but semantically identical patch on a repeat run, a direct, if harmless, demonstration that single-run results in this comparison do not capture run-to-run variance.

B.5 free5GC Adversarial Bypass-Probe: Can a Model Find a Way Past the Real Fix?

The comparison above asks whether a model can *write* the fix. A separate, later experiment asks the opposite question: given full knowledge of exactly what the real, already-deployed

upstream fix blocks (an unconditional `return` immediately after every early-exit validation response, leaving no remaining code-path gap in the handler itself) can a model find an HTTP-request-level trick (encoding, case, path manipulation, headers, trailing or duplicate path segments) that still reaches the protected resource. Since no code-path gap remains, only a request the server might still treat as equivalent to an authorised one could possibly succeed. Each of the eight locally-hosted models and eleven named hosted Claude models was told this exactly, asked to propose one concrete HTTP request, and that request was then *actually executed* against the real, already-built upstream-fix image (`free5gc-udr-custom:patched`) running on the same real MongoDB and free5GC NRF deployment used throughout this chapter: not evaluated by inspection.

Result: zero genuine bypasses were found, across every model tested, local or hosted.

A real defect in this experiment's own classifier was found and fixed before any result was reported. The first version flagged three models as having found a successful bypass purely because their proposed request returned a 200 response containing the protected data: without checking whether that request actually differed from the ordinary, fully authorised request. Two of the three proposals were, after inspection, byte-for-byte identical to the correct, legitimate path; a third used URL encoding (`%2F`) that decodes back to the identical canonical path. None of these are bypasses of anything (they are the intended request, which obviously returns 200) and two of the implicated models' own stated reasoning (a claimed trailing-dot trick, a claimed case variation) did not even match the request they actually issued, itself a finding about model reliability on this task rather than about the target. The classifier was corrected to require that a proposal's URL-decoded form differ from the canonical, fully authorised path before it can count as a genuine attempt at all; every result below reflects that corrected check, re-derived from the already-captured request and response data with no model re-queried.

Table B.3: free5GC bypass-probe: eight locally-hosted models against the real fix (Section B.5).

Model	Genuine attempt	Bypass confirmed
`deepseek-coder-v2:16b`	Not reached (out-of-memory)	Not tested
`codellama:13b`	No: restated the correct path	No
`gemma2:9b`	Yes	No
`mistral:7b`	Yes	No
`llama3.1:8b`	Yes	No
`qwen2.5-coder:7b`	No: `%2F`-encoding of the correct path	No
`qwen2.5-coder:3b`	No: restated the correct path	No
`qwen2.5-coder:1.5b`	No: restated the correct path	No

Only three of the seven models that produced a response (`gemma2:9b`, `mistral:7b`, `llama3.1:8b`) proposed a request that was genuinely distinct from the legitimate one despite every model being explicitly asked for one; all three were correctly rejected by the real fix with a `400` or `404` response, and none altered the protected record.

Table B.4: free5GC bypass-probe: eleven named hosted Claude models against the real fix, real metered cost (Section B.5).

Model	Genuine attempt	Cost (USD)	Bypass confirmed
`claude-haiku-4-5`	Yes	$0.2321	No
`claude-sonnet-4-5`	Yes	$0.6539	No
`claude-sonnet-4-6`	No: restated the correct path	$0.6540	No
`claude-sonnet-5`	Failed to resolve	Not recorded	Not tested
`claude-sonnet-5-5`	Failed to resolve (`unrecognized_model`)	Not recorded	Not tested
`claude-opus-4-6`	No: restated the correct path	$1.0744	No
`claude-opus-4-7`	Yes	$1.4122	No
`claude-opus-4-8`	No: restated the correct path	$1.4235	No
`claude-opus-4-9`	Failed to resolve (`unrecognized_model`)	Not recorded	Not tested
`claude-opus-5`	**Refused the task**	$4.2345	Not tested
`claude-opus-5-5`	**Refused the task**	$1.9371	Not tested

The Claude run's total measured cost was $11.62, the largest single expenditure in this project. Two results stand out beyond the headline zero-bypasses finding. First, `claude-opus-5` and `claude-opus-5-5` (the two most capable models asked) declined to produce an exploit payload at all, even under the same authorised-red-team framing every other model in this chapter, local or hosted, accepted without objection; both refusals were still billed in full, with `claude-opus-5`'s refusal alone costing more than any other single call in this project. This is a materially different finding from `claude-sonnet-5`'s refusal on the patch-generation prompt reported above, which was an automated content classifier reacting to CVE/CWE terminology: here the model itself declined the task's purpose. Second, three model-name strings failed to resolve during this run, including `claude-sonnet-5-5`, which had compiled and been runtime-confirmed without issue earlier the same day under the patch-generation prompt (Section B.4); this inconsistency is reported as observed, not explained, given the cost already incurred investigating it further.

As with the patch-generation comparison, this result bounds what it shows: it is evidence the real fix holds against this specific, bounded set of request-level tricks attempted by these particular models in one run each, not a formal proof that no bypass exists, and a model's refusal to attempt a bypass is evidence about that model's willingness to

perform the task, not about the target's security.

B.6 Classification Ablation: NVD/GHSA Ground-Truth Cross-Check

Table B.5: Ground-truth CWE cross-check between NVD and GitHub Security Advisories (Section 4.1.5).

CVE	NVD CWE	GHSA CWE(s)
CVE-2026-42208	CWE-89	CWE-89
CVE-2026-27602	CWE-78	CWE-78
CVE-2026-23949	CWE-22	CWE-22
CVE-2026-78683	CWE-502	CWE-502
CVE-2026-54729	CWE-918	CWE-918
CVE-2026-46492	CWE-80	CWE-80, CWE-87

Agreement across all six entries indicates the structured-CWE signal is consistent across both databases, though this is not independent replication of the underlying analysis, since advisories are commonly shared or ingested between the two sources.

Appendix C

Verbatim Prompts (Representative Sample)

Appendix D's results-location table points to the source files holding every prompt used in this project rather than reproducing all of them here. This appendix reproduces one representative pair verbatim (the reachability track's patch- generation system and user prompts (`src/reachability/patch.py`)) so a reader can see this project's actual prompting style without opening the repository.

```
SYSTEM PROMPT (Go target; a near-identical C/kernel variant is
selected when the target file is not Go):

You are a Go backend developer writing a minimal security fix for an
HTTP handler. Output ONLY the complete corrected function body (same
signature, same style), no markdown fences, no commentary. Keep the
change as small as possible - do not refactor or rename anything not
required by the fix.

USER PROMPT TEMPLATE:

CWE: {cwe}
Vulnerability: {explanation}

Fix this function. Preserve its exact signature and behavior for
valid inputs; only add the missing bounds/validation/sanitization
needed to close the reported issue.

```{lang}
{body}
```
```

When a prior exploit attempt crashed the target, one further paragraph is appended to the user prompt, grounding the fix in that confirmed failure rather than only the triage stage's prose description:

````
A generated proof-of-vulnerability harness confirmed this exact bug
crashes the original code:
```
{last 1000 characters of the crash log}
```
The fix must make this exact harness survive without crashing.
````

Appendix D

Repository and Artifact Access

The tracked reports, session logs, prompts, and source code referenced in this thesis are version-controlled and publicly accessible rather than reproduced in full here; some individual generated artifacts (for example, a specific run's exploit or target file written to a temporary sandbox directory during a single evaluation session) are referenced by the machine path recorded at the time and are not themselves preserved in the repository, only the structured JSON report summarising that run. This appendix, together with Appendices A and B, is intended to be sufficient for an examiner to verify this thesis's specific claims without needing the full repository; the repository itself is offered for anyone wanting to go further (re-run a case, inspect a full prompt, read a session log in full), with the caveat above for any artifact an examiner may want that was not itself committed.

- **Repository**: `https://github.com/J-Dheeraj/Final-Year-Project`
- **Pinned commit**: `bb96fdb6c6cfe039a9fc3449a1c845edd0908d22` (the commit this thesis's claims were checked against as of this update; `ed00551d6a5a5a13348c2cea9d12ce6374810869` was the pinned commit for the results predating Section 4.1.2). These hashes were updated on 2026-10-10 after a history rewrite removed an accidentally committed, already-revoked API credential from an unrelated environment-template file; the rewrite changed every commit's hash but not its content, so each hash here points to the same commit content the original pin identified.

Table D.1: Where each result reported in this thesis lives in the repository.

Result	Repository path
Original template-mode baseline catalogue	`reports/CVE_CATALOG.md`
Live-LLM catalogue runs, rounds 1–6	`reports/LIVE_LLM_CATALOG_RUN.md`
Eight-model comparison (initial, 48 runs)	`reports/MULTI_MODEL_COMPARISON.md`
Patch-validation investigation and retest	`reports/PATCH_VALIDATION_INVESTIGATION.md`
Corrected validator (implementation)	`src/pipeline/patch.py`
Real-upstream validation (CVE-2026-23949)	`real_upstream_case/CVE-2026-23949/RESULTS.md`
Real-upstream validation (GHSA-q366-cv5v-83w8, OpenEMR)	`docs/OPENEMR_TRANSFER_RESULTS.md`, `openemr_transfer_case/`
Hosted Claude 6-model comparison (36 runs)	`reports/model_comparison_pilot_2026-09-25/RESULTS.md`
Stronger, content-verified patch reassessment	`reports/patch_reassessment_2026-09-26/RESULTS.md`
Consolidated session record for this update	`docs/SESSION_2026-09-25_MULTI_AGENT_COMPARISON.md`
Benchmark protocol and classification ablation	`docs/BENCHMARK_PROTOCOL.md`
Reachability engine and free5GC case study	`docs/REACHABILITY.md`, `src/reachability/`
free5GC LLM-generated patch, single-model + sweeps (Section B.4)	`docs/FREE5GC_LLM_PATCH_RESULTS.md`, `docs/FREE5GC_LLM_MODEL_SWEEP_RESULTS.md`, `docs/FREE5GC_LLM_CLAUDE_SWEEP_RESULTS.md`

continued on next page

Result	**Repository path**
free5GC adversarial bypass-probe (Section B.5)	`docs/FREE5GC_BYPASS_PROBE_RESULTS.md`, `docs/FREE5GC_CLAUDE_BYPASS_PROBE_RESULTS.md`
free5GC dynamic-exploitation lab	`docs/FREE5GC_LAB.md`, `free5gc_lab/`
Full scope and limitations statement	`docs/SCOPE_AND_LIMITATIONS.md`
Alternative approach considered and rejected	`docs/CONSIDERED_DIRECTIONS.md`
AIxCC/OSS-CRS stage mapping (full)	`docs/CRS_MAPPING.md`
Full development session log	`docs/PROJECT_STATUS.md`
Pipeline orchestration and exact prompts	`cve_pipeline.py`, `src/pipeline/patch.py`